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9 June 2016

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MANUAL

GEOINT Functional Manager Standards Assessment (GFMSA) Program

A program for maturing consistent and conforming application of GEOINT standards for interoperable operations within the NSG

Originating Office:
National Geospatial-Intelligence Agency
Chief Information Officer and Information Technology Services

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FOREWORD

The GEOINT Functional Manager Standards Assessment (GFMSA) Program was established by the GEOINT Functional Manager (GEOINT FM) to promote a GEOINT standards-based, interoperable working environment across the National System for Geospatial Intelligence (NSG). The Program is also intended to ensure that GEOINT requirements are considered throughout the acquisition procedures and testing activities (Reference j). This program addresses GEOINT standards approved by the Intelligence Community (IC) and Department of Defense (DoD) Joint Enterprise Standards Committee (JESC) (Reference n z).

The GFMSA program establishes processes that provide program and project managers an authoritative means to confirm that interoperable capabilities are delivered that conform to GEOINT standards. Starting those processes early in the requirements phase of the acquisition life cycle provides Program Managers (PMs) with a higher level of confidence for an interoperable outcome. As a result, PMs can confirm that their system produces, provides, or uses GEOINT in conformance with mandated standards.

These processes are built around an assessment approach that adds significant fidelity and confidence to the GEOINT functional aspect of a program or project. This reduces the research (time and cost), development, and maintenance work a program must conduct to understand application of GEOINT standards and the interoperability objectives and operational capabilities that need to be realized across the NSG.

Bottom line: Information contained in this Manual, and on the GFMSA resource sites, will help managers and sponsors improve their GEOINT requirement statements and acquisition development processes. By following the processes established in this Manual the NSG, including managers and sponsors, will be able to answer these key questions: 1) What interoperable capabilities do I need to have as a result of my program/system requirements; and 2) How will I know when they have been delivered?

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CHANGE LOG

Change Type & Number	Change Date	Revised By	Pages Affected	Change Description
Admin 1	12/7/2018	TAES:kvf	All	JCIDS Manual update required changing “N-R KPP” to “Net-Ready Performance Attribute”. Incorporate recent changes updates caused by changes to the GFMSA Scorecard, and to improve usability of the document. Update to the list of references necessitated by changes to the referenced documents. Language clarification and removal of unused information.

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1. PURPOSE.

a. This GFMSA Manual implements policy, defines responsibilities, and provides processes for assessing and asserting conformance to GEOINT standards in accordance with NSGD 3201, "The Geospatial Intelligence (GEOINT) Functional Manager Standards Assessment (GFMSA) Program" (Reference ~~j u~~).

b. This Manual focuses on the capabilities enabled by GEOINT data and GEOINT web services.

2. APPLICABILITY. This Manual applies (to the extent described in NSGD 3201 (Reference ~~j u~~)) to elements of the IC, DoD, non-DoD/IC Federal agency members of the NSG, international partners, state/local municipalities, and Native American tribal organizations that are responsible for the operation, acquisition and/or development of systems and applications which collect, procure, produce, serve, exchange, or use GEOINT data. ~~The GFMSA Self-Assessment Guide contains detailed instructions and procedures for self-assessments. This document is available on the GFMSA resource sites listed in Enclosure 2, paragraph 4.~~

3. REFERENCES. See Enclosure 1.

4. ACRONYMS AND DEFINITIONS. See Glossary, Part 1 and Part 2.

5. RESPONSIBILITIES. See Enclosure 2.

6. PROCESSES. See Enclosure 3.

7. ASSESSMENT OBJECTIVES and CRITERIA. See Enclosure 4.

8. ADMINISTRATION. The GFMSA Manager(s) is/are responsible for administration and maintenance of this Manual.

9. RELEASABILITY. This document is available on the ~~GFMSA resource sites listed in Enclosure 2, paragraph 4, and on~~ the NSG Standards Registry, (Reference ~~p ee~~), ~~and from the GFMSA resource sites listed in Enclosure 2, paragraph 4.~~

10. EFFECTIVE DATE. This Manual is effective immediately.

Dustin J. Gard-Weiss
Chairman, National Geospatial Intelligence Committee

Enclosures

1. References
2. Responsibilities

- 3. Processes
- 4. Assessment Objectives and Criteria
- 5. Glossary

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ENCLOSURE 2**RESPONSIBILITIES**

1. INTRODUCTION. The GFMSA program is focused on ensuring consistent selection of and conformance to GEOINT standards by systems that collect, produce, provide, serve, exchange or use GEOINT. Conformance to NSG-adopted GEOINT standards is critical to interoperability and the ability of the NSG enterprise to effectively and efficiently collect, procure, produce, serve, use or exchange GEOINT data. The standards are those adopted by the JESC (Reference ~~n z~~), and found in the DoD Information Technology Standards Registry (DISR) or the IC Enterprise Standards Baseline (IC ESB).

The DoD, IC, and other NSG members/partners have different acquisition and development process and milestone models. Some developments use various forms of the agile development methodology. To permit agility in assessment processes, in-progress self-assessments are organized to align with the generic acquisition activities identified in NSGD 3201 (Reference ~~j u~~) and listed below:

- a. Requirements definition and validation
- b. Standards selection and system design
- c. Standards implementation and development testing
- d. Operational and interoperability testing

2. INFORMATION TECHNOLOGY (IT)/NATIONAL SECURITY SYSTEMS (NSS)
SPONSORS AND PROGRAM MANAGERS

a. Verify and ensure certified requirements, to include the Net-Ready ~~Performance Attribute Key Performance Parameter (N-R KPP)~~ and ~~Intelligence Supportability~~, that address GEOINT standards conformance and interoperability requirements for effective mission capabilities and have been developed and certified through DoD Components per NSGD 3201 (Reference ~~j u~~) in order to satisfy standards prescribed by the GEOINT FM In Accordance With (IAW) DoDI 8330.01 (Reference ~~d k~~) and the ~~JCIDS Manual (Reference f)~~.

b. Ensure mandated standards that provide solutions to the validated/certified requirements are implemented to better realize NSG-wide interoperability or submit a waiver request IAW the Joint Interoperability Test Command (JITC) Interoperability Process Guide (Reference ~~o dd~~).

c. Coordinate with the GFMSA Manager(s) in the review of IT/NSS developmental, operational, and interoperability test plans to gain as much GEOINT-related assessment

data from those events as possible.

d. Self-determine which opportunities in their schedule best align with the generic activities defined in NSGD 3201 for conducting and reporting in-progress self-assessments to the GFMSA Manager(s).

e. Self-assess the degree to which GEOINT data and web services designs conform to GEOINT standards, meet GFMSA objectives and criteria, and are on-track for enabling system capability requirements.

f. Self-assess the degree to which implemented GEOINT data and/or services conform to standards as specified by the system design, meet GFMSA objectives (Enclosure 4), and are on-track for enabling system capability requirements.

g. Self-assess the degree to which GEOINT-related capability and **net-ready performance attribute N-R-KPP** measures of effectiveness and performance thresholds are met.

h. Inform the GFMSA Manager(s) about major gaps or shortfalls to achieving GEOINT standards conformance and meeting effectiveness and performance thresholds, and plans for remediation.

i. In coordination with the GFMSA Manager(s), submit documentation of standards conformance and operational validation to the Interoperability Certification Authority. (Reference **d k**).

j. Require that IT programs be adequately funded to execute the interoperability functions specified in this Manual. (Reference **d k**).

3. GFMSA MANAGER(S)

a. Confirm the integration of GEOINT standards in systems that collect, procure, produce, serve, exchange or use GEOINT by taking part in reviews of GEOINT requirements, and by verifying the **net-ready performance attribute N-R-KPP and Intelligence Supportability** are ~~is~~ adequately defined for GEOINT.

b. Review IT test strategies and developmental and operational test plans to verify that all GEOINT-related requirements are addressed.

c. Review self-assessment substantiating materials to verify that GEOINT requirements are satisfied in conformance with NSG-wide GEOINT standards.

d. Coordinate with **net-ready performance attribute N-R-KPP, Intelligence Supportability**, and interoperability certification authorities to ensure that GEOINT **net-ready performance attribute N-R-KPP** and interoperability test and evaluation criteria,

measures, and requirements are considered before those authorities grant certifications.

e. Represent the GEOINT FM on the Interoperability Steering Group (ISG) (Reference ~~d k~~). The ISG adjudicates Interoperability Certification requests, but may waive the requirement and issue an Interim Certification To Operate (ICTO) on DoD networks.

4. POINTS OF CONTACT

A program or project manager can obtain technical assistance with developing and implementing systems or products with GEOINT capabilities by contacting the GFMSA Manager(s).

Email: GFMSA_Team@nga.mil
GFMSA_Team@nga.smil.mil
~~GFMSA_Team@coe.ic.gov~~
~~GFMSA_Team@nga.ic.gov~~

GFMSA Resource Sites:

~~<https://intelshare.intelink.gov/sites/gfmsa> *~~
~~<https://intelshare.intelink.gov/sites/gfmsa/default.aspx>~~
~~<https://intelshare.intelink.sgov.gov/sites/gfmsa> **~~
~~<https://intelshare.intelink.sgov.gov/sites/gfmsa/default.aspx>~~
~~<https://intelshare.intelink.ic.gov/sites/gfmsa>~~
~~<https://intelshare.intelink.ic.gov/sites/gfmsa/default.aspx>~~

*** Requires .mil or .gov domain with CAC or PIV to access. ~~To obtain a copy, send an e-mail to: GFMSA_Team@nga.mil~~**

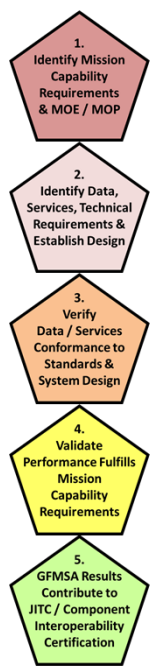
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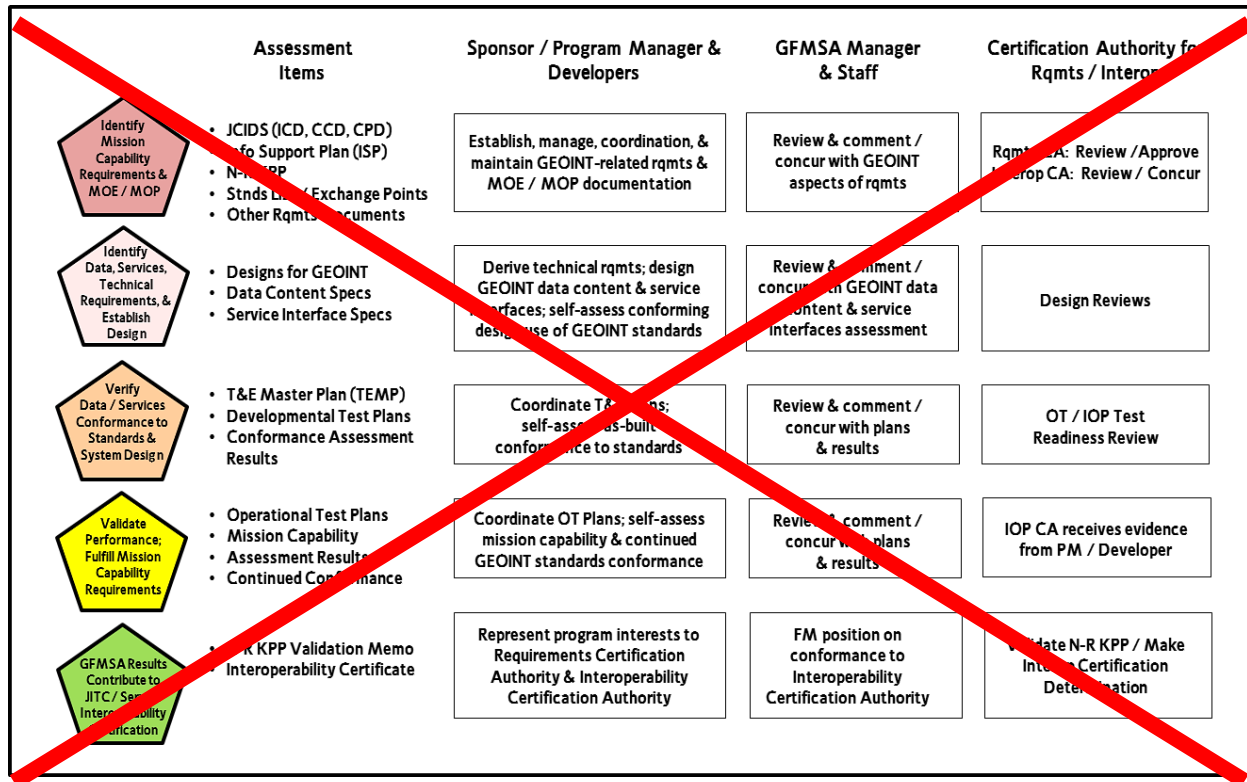
ENCLOSURE 3**PROCESSES**

1. **INTRODUCTION.** The GFMSA program provides processes for program managers to obtain easily-recognized, community-wide acknowledgement that an IT component has been assessed in a credible manner and found to meet the standards conformance criteria for interoperability as set forth herein by the NSG community. Successful satisfaction of the standards conformance assessment and assertion process will optimize potential for a system to meet its GEOINT-related operational performance objectives while attaining interoperability certification (Reference **d k**). A general overview of items to be self-assessed and organizational responsibilities is depicted in Figure 1. See Section 3, Path to Standards-Conforming Interoperability, for details.

Figure 1. Process Documents and Actors



	Assessment Items	Sponsor / Program Manager & Developers	GFMSA Manager & Staff	Certification Authority
1. Identify Mission Capability Requirements & MOE / MOP	- JCIDS (ICD, CDD, CPD) - ICCR (SOC) - Info Support Plan (ISP) - Net Ready Performance Attribute - StdV-1 & SV-6 or SvcV-6	Establish, manage, coordinate, and maintain GEOINT-related requirements & MOE / MOP documentation	Review and comment on GEOINT aspects of requirements	Requirements, Interoperability, and Intelligence
2. Identify Data, Services, Technical Requirements & Establish Design	- Designs for GEOINT - Data Content Specs - Service Interface Specs	Derive technical requirements. Design GEOINT data content and service interfaces. Self-assess conforming design use of GEOINT standards	Review and comment on GEOINT data content and service interfaces assessment	Design Reviews
3. Verify Data / Services Conformance to Standards & System Design	- T&E Master Plan (TEMP) - Developmental Test Plans - Conformance Assessment Results	Coordinate T&E Plans. Self-assess as-built conformance to standards	Review and comment on plans and results	OT / Interoperability Test Readiness Review
4. Validate Performance Fulfills Mission Capability Requirements	- Operational Test Plans - Mission Capability Assessment Results - Continued Conformance	Coordinate OT Plans. Self-assess mission capability & continued GEOINT standards conformance	Review and comment on plans and results	Interoperability CA receives evidence from PM / Developer
5. GFMSA Results Contribute to JIJC / Component Interoperability Certification	Interoperability Certificate	Represent program interests to Requirements Certification Authority & Interoperability Certification Authority (CA)	GFM position on conformance to Interoperability Certification Authority	Validate Net-Ready Performance Attribute / Make Interoperability Certification Determination



2. GFMSA APPROACH. The GFMSA approach includes:

a. Aids for preparation of GEOINT-related standards conformance and performance requirements, Measures of Effectiveness (MOEs), Measures of Performance (MOPs), and the **net-ready performance attribute - Net-Ready Key Performance Parameter (N-R KPP)**.

b. GEOINT Standards Working Group (GWG)-adopted standardization objectives and criteria for GEOINT data and web services to be deemed Visible, Accessible, Understandable, Trusted, and Interoperable (VAUTI), (Reference **c j**). See Section 6. GFMSA Objectives – GEOINT Data and Services.

c. Credible, yet flexible, risk-reduction processes for programs to self-assess and assert GEOINT standards conformance.

d. Identification of items to be assessed for standards conformance.

e. Information about availability and capabilities of GEOINT-related means of test and test services.

f. Guidance for meeting GEOINT-related processes and interoperability responsibilities set forth by (Reference **d k**) to:

(1) Confirm integration of GEOINT standards in GEOINT-related Information Support Plans (ISPs).

(2) Verify the ~~net-ready performance attribute N-R-KPP~~ is adequately defined to address GEOINT requirements.

(3) Verify that GEOINT requirements are addressed in IT developmental, operational, and interoperability test plans.

(4) Review test results to verify that GEOINT-related requirements are satisfied.

(5) Ensure that GEOINT interoperability test and evaluation criteria, measures, and requirements are fulfilled before certification authorities grant interoperability certifications.

g. In-progress status reporting on the state of meeting GEOINT standards conformance and interoperability objectives during the system development and testing cycle.

h. The GFMSA Resource Sites (~~Nonclassified Internet Protocol Router Network (NIPRNET) Sensitive But Unclassified (SBU)~~/Secret Internet Protocol Router Network (SIPRNET)/Joint Worldwide Intelligence Communications System (JWICS)) serve as a central location for the GEOINT community to mutually share information about available test and evaluation (T&E) infrastructure, test tools, and test service providers.

3. PATH TO STANDARDS-CONFORMING INTEROPERABILITY

a. Overview. The GFMSA process defines five focus areas to assess how well system developments are progressing towards meeting the VAUTI objectives and criteria for asserting standards conforming interoperability as described in Enclosure 4. The GFMSA cycle, depicted in Figure 2, can be an overlapping, repetitive, and/or regressive process as conditions change. The focus areas are:

(1) Identify mission capability requirements and associated Measures of Effectiveness (MOE) and Measures of Performance (MOP).

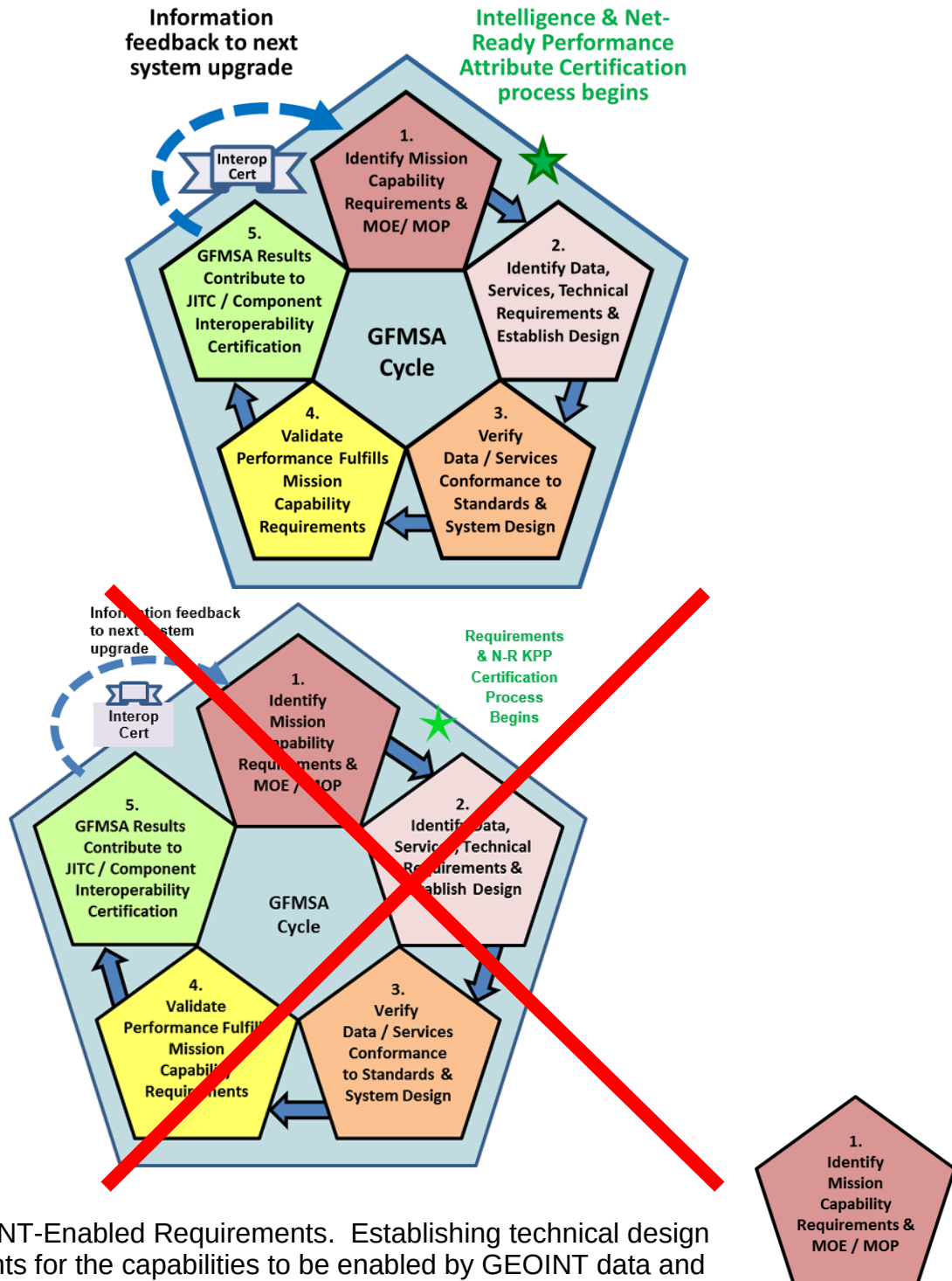
(2) Identify data, service, and technical requirements; and establish system data and services design in conformance with applicable GEOINT standards.

(3) Verify that data and services conform to both the GEOINT standards and the system data and services design for implementing those standards.

(4) Validate that the performance of the conforming design, data, and services fulfill the approved mission capability, effectiveness, and performance requirements established in focus area 1.

(5) Submit evidence substantiating GFMSA objectives and criteria are met to the GEOINT FM for endorsement to the interoperability certification authority. GFMSA results contribute to an interoperability certification determination.

Figure 2. GFMSA Cycle



b. GEOINT-Enabled Requirements. Establishing technical design requirements for the capabilities to be enabled by GEOINT data and

services is a critical activity, and component/system sponsors must resolve any requirements/capabilities issues with the appropriate Milestone Decision Authority (MDA). The GFMSA process enables identification of GEOINT requirements in the Defense and IC requirements and acquisition processes. Successful implementation of validated/certified requirements for GEOINT data and services is confirmed by GEOINT-aware T&E activities. These T&E activities reflect the requirements of the following documents:

~~(1) CJCSI 3170.01I, "Joint Capabilities Integration and Development System (JCIDS)" (Reference f o). The capabilities development process has been strengthened with the publication of CJCSI 3170.01I. The JCIDS supports the Joint Staff and the Joint Requirements Oversight Council (JROC) in identifying, assessing, and prioritizing joint military capability needs. As prescribed by the JCIDS process, technical assessment is needed to ensure IT and NSS capability and requirements documents are specified in measurable and testable forms. Appendix E to Enclosure D of the JCIDS Manual (Reference g p) provides details for the Net-Ready Key Performance Parameter which is a key part of the GFMSA process.~~

~~(1 2) Intelligence Community Policy Guidance Number 801.1, "Acquisition," 12 July 2007, as amended, 29 January 2015 (Reference e n). This IC acquisition document includes the Intelligence Community Acquisition Model (ICAM) which defines the acquisition cycle to be used by members of the IC. Those NSG members who are also members of the IC will use this document to guide their acquisition processes. Please note that use of the ICAM does not relieve those members that are also DoD Combat Support Agencies of the requirements found in Reference d k.~~

~~(2 3) CJCSI 5123.01H G, "Charter of the Joint Requirements Oversight Council (JROC) and Implementation of the Joint Capabilities Integration and Development System (JCIDS)" (Reference g q). This Instruction canceled both N-R KPP Manual, "Net Ready Key Performance Parameter (N-R KPP)" and CJCSI 3312.01B, "Joint Military Intelligence Requirements Certification (IRC)," and now establishes the policies and procedures for both under the JCIDS. This change makes the document directly GEOINT-aware to better ensure that GFMSA objectives, associated criteria, and measurable performance parameters are addressed.~~

~~(3 4) The Appendix E to Enclosure D of the JCIDS Manual (31 August 2018 12 February 2015), "Content Guide for the Net-Ready Key Performance Parameter (N-R KPP)" (Reference f p). Testability information should be contained in the Joint Interoperability or Intelligence Supportability sections as well as in the applicable KPPs and performance attributes. This Manual defines procedures to develop the N-R KPP and its certification requirement for all IT and NSS to prepare for the review of the N-R KPP and architecture data, as applicable, in all JCIDS documents and ISPs.~~

~~(a) The N-R KPP is one of the six mandatory KPPs that must be considered.~~ The intent of the net-ready performance attribute N-R KPP is to ensure new systems fit into the existing architectures and infrastructure to the maximum extent practicable.

(b) For the ~~net-ready performance attribute N-R-KPP~~, system sponsors identify the capability requirement for each mission or function, and at least one measurable performance attribute that is critical to the system. Threshold values are then developed based on what is achievable through the current state of technology as a minimum; and objective values are based on a goal for the end-state of the system.

(c) The JCIDS Manual provides example Key Performance Parameter (KPP) attributes ~~and performance attributes~~ for the various Joint Capability Areas (JCAs). As examples, Table 1 lists JCIDS Manual extracts of ~~performance attributes and intelligence supportability~~ for data (e.g., GEOINT) dissemination, analysis, prediction, and production services. Information contained in this Manual, and on the GFMSA Resource Site, helps program/project managers and component/system sponsors bring improved GEOINT awareness to the requirements and acquisition development processes.

Table 1. Exemplar Intelligence Supportability and Net-Ready Performance Attributes

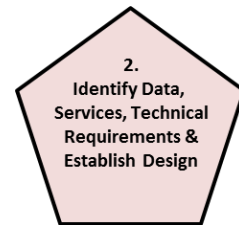
Intelligence Supportability				N-R	Attributes
Targeting	Intelligence Interoperability	Intelligence Mission Data	Space Intelligence	Interoperability	
	X				Ability to authenticate users/machines and make authorization decisions for their access to information
				X	Ability to discover and retrieve information for all appropriate data sources
		X	X		Ability to hold track (time and type of target)
	X	X			Ability to integrate, evaluate, interpret, and predict knowledge and information from available sources to develop intelligence and forecast the future state
				X	Ability to securely transmit data from collector through a media link to a processing site
				X	Ability to support data relay with adequate capacity, continuity, and reliability
		X	X		Contiguous area (wide and narrow field of view)
	X	X			Effective range to target for onboard sensors under differing weather conditions
X		X			Endurance once on target
X	X	X	X		Geolocation accuracy and precision
		X			Image quality
X		X			Minimum detectable velocities
	X				Number and type of data sources able to be fused together
	X	X			Resolution in National Imagery Interpretability Rating Scale (NIIRS) or Ground Sample Distance (GSD)
	X	X	X		Revisit rates or intervals
	X		X	X	Sensor tasking
X		X			Simultaneity
	X	X			Spectrum covered by sensor collection
		X			Synoptic area coverage
X	X	X			Time on target
	X			X	Time to report; once data is collected, time to requested user
		X	X	X	Time to target or re-target sensors

Table 1. Exemplar N-R KPP Attributes

Contiguous area (wide and narrow field of view)
Simultaneity
Synoptic area coverage
Effective range to target for onboard sensors under differing weather conditions
Time on target

Endurance once on target
Revisit rates or intervals
Time to target or re-target sensors
Time to report; once data is collected, time to requested user
Bandwidth range collected against
Geolocation accuracy and precision
Resolution in National Imagery Interpretability Rating Scale (NIIRS) or Ground Sample Distance (GSD)
Spectrum covered by sensor collection
Minimum detectable velocities
Ability to hold track (time and type of target)
Image quality
Image interpretability
Ability to discover and retrieve information for all appropriate data sources
Ability to authenticate users/machines and make authorization decisions for their access to information
Ability to securely transmit data from collector through a media link to a processing site
Ability to support the data relay with adequate capacity, continuity, and reliability
Ability to integrate, evaluate, interpret, and predict knowledge and information from available sources to develop intelligence and forecast the future state
Number and type of data sources able to be fused together
Time spent data mining vs. time spent performing analysis, prediction, and production

c. GEOINT Awareness by Design. Mission capability requirements are stated in operational terms typically expressed as required warfighting capabilities, or as a technological development that presents an opportunity to enhance military capabilities. Translating mission capability requirements into an implementable system design for which GEOINT data and services are in conformance with applicable GEOINT standards is critical for achieving mission capabilities in an enterprise such as that supporting the NSG. Assessing the design (GEOINT data and service specifications) for standards conformance promotes program confidence that the data and service design solution will satisfy the desired mission capabilities.



**You can't test a system into standards conformance.
It must be conforming by design.**

d. GEOINT Standards and Development Timelines. A common design and development decision point is when to freeze the design and the versions of GEOINT standards. GFMSA objectives and criteria for data and services may be satisfied using older versions of GEOINT standards when mission requirements are met under one or more of the following circumstances:

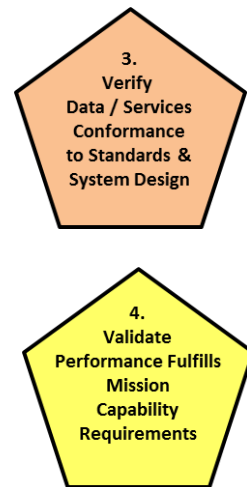
(1) The design-selected GEOINT standards were cited in the DISR/IC ESB baseline at the time the data and/or services specification design was frozen.

(2) The design-selected GEOINT standards are cited in a configuration managed product specification, services interface control document, implementation profile, or similar design documentation current at the time the data content and/or services design was frozen.

(3) The design-selected GEOINT standards remain generally supported within the NSG enterprise, and are supported by intended use applications and users in supporting mission effective capabilities.

e. GEOINT-Aware Testing and Assessment Reporting

(1) While the terms “verification and validation” are often used together, there is an important distinction depicted in the GFMSA cycle to first verify conformance with the design and its use of GEOINT standards, and then validate that the conforming data and services fulfill the mission capability requirements. Verification of standards conformance is principally conducted during the developmental testing phase. Verification should be substantially completed as one of the entry conditions considered at the test-readiness-review prior to proceeding to validation test activities during the operational test phase. Test planning commences during the early stages of the requirements definition and acquisition processes. (Enclosure 4, Section 2), Test and Evaluation Planning, Conduct, and Reporting).



Early GEOINT considerations will decrease overall program costs, maximize a program's GEOINT capability, and decrease GEOINT-related technical and schedule risk.

(2) Program/Project Managers conduct in-progress assessments prior to Milestone (MS) or major program reviews to ensure that all appropriate GEOINT capabilities have been accounted for, and design considerations have been made. The GFMSA process tracks the in-progress degree of satisfying GFMSA objectives and criteria for those programs on the path to

standards conformance-based interoperability at these significant requirements and acquisition decision points and milestones. Early GEOINT considerations will decrease overall program costs, maximize a program's GEOINT capability, and decrease GEOINT-related technical and schedule risk. Further, in-progress reviews increase the likelihood of meeting GFMSA objectives and criteria.

(3) Typically, programs/projects conduct and report **five focus area four** in-progress assessments of GEOINT standards conformance in consonance with the system's acquisition **phase stages** corresponding to:

- (a) Requirements definition and validation
- (b) Standards selection and system design

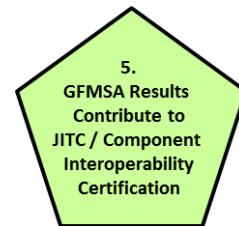
(c) Standards implementation and development testing

(d) Operational and interoperability testing

(e) Interoperability Certification

f. Substantiation

(1) Program/Project managers conduct self-assessments, and submit evidence substantiating that GFMSA objectives and criteria have been successfully met. Sources of substantiating evidence include results from documentation reviews, analysis, demonstrations, evaluations, examinations, inspections, tests, questionnaires, surveys, conformance registers, collateral sources, and similar activities that confirm conforming implementation of GEOINT standards in a manner enabling interoperability across the NSG enterprise. Refer to the online GFMSA Resource Site (Enclosure 2, Section 4, Points of Contact for the link) for self-assessment tools and instructions about making evidence submissions.



(2) Configuration Management (CM) plays a significant role in the GFMSA process. GFMSA objectives and criteria must be reassessed when version changes occur to data and/or services supporting mission capability requirements. The assessment rigor and substantiation evidence will be commensurate with the nature and extent of the changes to the earlier assessed version. Program/ Project Managers, in coordination with the GEOINT FM staff, submit evidence of GEOINT standards conformance and operational validation to the Interoperability Certification Authority (Reference **d k**).

g. In-Progress Status Indicators. In-progress status indicators are organized by GFMSA Cycle focus areas as described in 3.a (1)-(5) and depicted in Figure 2, above. Those performing the in-progress assessment assign progress indicator values in consonance with the system's acquisition **phase stages** at time of assessment. The indicator values are listed in Table 2, below. High level questions for those focus areas, by number, are:

(1) Focus Area 1. Identify Mission Capability Requirements and MOE/MOP.

(a) Validated/Certified requirements and **net-ready performance attribute N-R KPP** thresholds and objectives address GEOINT standards conformance and interoperability requirements sufficient to enable effective mission capabilities.

(b) Acquisition documents require implementation of GEOINT standards with means to deliver substantiating evidence of GEOINT standards conformance and interoperability.

(2) Focus Area 2. Identify Data, Services and Technical Requirements, and Establish Design.

(a) GEOINT data design conforms to GEOINT standards, meets VAUTI objectives and is on-track for enabling system mission.

(b) GEOINT services design conforms to GEOINT standards, meets VAUTI objectives and is on-track for enabling system mission.

(3) Focus Area 3. Verify Data/Services Conformance to Standards and System Design.

(a) GEOINT data conforms to standards as specified by the system design, meets VAUTI objectives and is on-track for enabling system capability requirements.

(b) GEOINT services conform to standards as specified by the system design, meet VAUTI objectives and are on-track for enabling system capability requirements.

(4) Focus Area 4. Validate Performance Fulfills Mission Capability Requirements.

(a) GEOINT-related capability effectiveness and performance thresholds are met.

(b) GEOINT-related ~~net-ready performance attribute N-R-KPP~~ effectiveness and performance thresholds are met.

(5) Focus Area 5. Submit GFMSA Evidence to Interoperability Certification Authority (Reference ~~d k~~).

(a) GEOINT FM-approved evidence of GEOINT standards conformance submitted to Interoperability Certification Authority.

(b) System is interoperability certified.

Table 2. In-Progress Indicator

Score	Assessment	Assessment Status Description
80%-100%	Assessed; On Track	High probability of standards conformance: metric substantially met for the in-progress phase under evaluation.

60% - 79%	Assessed; On Track with Shortfalls	Working toward GEOINT standards conformance: metric progressing for the in-progress phase under evaluation, but not yet met.
0% - 59%	Assessed; Not On Track	Low probability of standards conformance: inadequate (little/no) progress made toward meeting the metric for the in-progress phase.

0	Not Assessed; Status Unknown	No activity/attempt has been made towards evaluating the metric during the in-progress stage under evaluation.
1	Assessed; Not On Track	Inadequate (little/no) progress made toward meeting the metric, or attempts to meet the metric have failed during the in-progress stage under evaluation.
2	Assessed; Marginally On Track	Metric marginally progressing for the in-progress stage under evaluation, but not yet met.
3	Assessed; On Track with Shortfalls	Metric progressing for the in-progress phase under evaluation, but not yet met.
4	Assessed; On Track with Minor Shortfalls	Metric actively progressing for the in-progress stage under evaluation, but not yet met.
5	Assessed; On Track	Metric substantially met for the in-progress stage under evaluation.

h. Overall **Assessment Status Levels**. In recognition that many programs are already in development or fielded (are not new starts), three levels of overall GFMSA status are defined to characterize continuing progress toward achieving GFMSA objectives. The overall **assessment status level** is derived from **the scores the in-progress status indicators** identified in Table 2. **The scores are expressed as rounded integers as they are only estimates. The Summary Tab of the GFMSA Scorecard states:**

Estimate/Likelihood of GEOINT Standards Conformance*

**This is not a guarantee, but an indicator of conformance.*

(1) Green (80% - 100%). Green status is for programs/projects that substantially meet, or are on the path to substantially meet, the GFMSA objectives and criteria in context of the acquisition ~~phase stages~~ or development phase for which the assessment is conducted.

(2) Yellow (60% - 79%). Yellow status is for programs/projects that are not quite on the path to meet GFMSA objectives and criteria, but the deficiencies pose low operational risk and resolution will occur in a planned acquisition ~~phase stages~~, development phase, or incremental upgrade to the system.

(3) Red (0% - 59%). Red status is for programs/projects that are not on a path that will meet GFMSA objectives and criteria (e.g., deficiencies pose high operational risk with no plan or expectation of resolution).

i. Acknowledgement. Findings, results, and status are incorporated into the GEOINT Manager's Annual Report to inform the SecDef, the DNI, and the NSG on the state of GEOINT standards conformance, and achievement of NSG-wide interoperability by assessed IT. Components/systems assessed to GFMSA objectives and criteria are ~~available to the submitting program in the Program Document Library on their Program SharePoint site which is a sub-site of cited on applicable registers within the GFMSA Registry on the GFMSA SharePoint Resource Site.~~

j. GFMSA ~~SharePoint Resource~~ Site. The online GFMSA ~~SharePoint Resource~~ Site provides a centralized location for programs/projects to submit GFMSA substantiation materials, track in-progress status, and a ~~document library for self-assessment feedback by use of the In-Progress Review registry~~ to acknowledge successful completion of GFMSA objectives. Individual ~~sub-sites registers~~ are established to reflect the general categories of NSG components/systems for which NSG objectives and criteria have been established.

This site also contains informational resources in support of the GFMSA process. These resources include:

- (1) An overview description of the GFMSA procedures.
- (2) A description of the GFMSA objectives and criteria applicable to each ~~sub-site Register~~.
- (3) An area with information regarding the "State of Testability" for GEOINT standards and known test services providers.
- (4) Information about technical services that are made available to program/project offices.
- (5) GFMSA-specific documentation with a content synopsis for each document.

- (6) An area to post draft GFMSA documentation for ~~community~~ review.
- (7) An announcement and bulletin area for highlighting ongoing and planned events.
- (8) A means for users to ascertain content has changed since last visiting the site.
- k. Waiver Process. The GFMSA program honors waivers approved by standards and interoperability waiver decision authorities. GFMSA status is based on meeting the objectives and criteria stated herein (see paragraph 3.h, **Overall Assessment Status Levels**). For the DoD, the process for managing and granting waivers to standards is addressed in DoDI 8310.01, Information Technology Standards in the DoD. The process for managing and granting waivers to interoperability certification is addressed in DoDI 8330.01, Interoperability of Information Technology (IT), including National Security Systems (NSS). A similar process takes place for the IC (**Reference r**) and other NSG member organizations. The GEOINT FM coordinates with and advises NSG member organizations on matters involving requested waivers to GEOINT standards, and represents NSG enterprise interests to decision-making authorities when considering waiver requests.
- l. Dispute Resolution. Disputes concerning the provisions of this Manual, including applicability to IT Systems, conduct of the GFMSA process, assessment results, and/or agreement between PMs and GFMSA Manager(s) regarding which standards should apply to specific programs should be first worked between PMs and GFMSA Manager(s), and if needed, elevated to Component Heads and the GEOINT FM. If no resolution is reached between Component Heads and the GEOINT FM, per NSGD 3201 (Reference **j** ~~u~~) disputes are referred to DNI or USD(I) for resolution.

4. BENEFITS

- a. Program and project managers executing GFMSA benefit from:
 - (1) Technical consultation for the identification and implementation of GEOINT standards necessary for the system.
 - (2) Assistance in preparing GEOINT-related inputs for Test and Evaluation Strategies, Test and Evaluation Master Plans (TEMPS), and similar test planning documentation.
 - (3) Identification of GEOINT-aware test service providers.
 - (4) NSG-vetted test criteria and measures for GEOINT.

- (5) Improved GEOINT test and evaluation cost projections for budget planning.
 - (6) Sample data exemplifying proper application of GEOINT data standards.
 - (7) Access to sample service interfaces exemplifying proper application of GEOINT interface standards.
 - (8) In-progress assessment and review services provided as part of development process to minimize risk of inadvertent misapplication of GEOINT standards.
 - (9) Reduced technical and schedule risk achieved through ready access to GEOINT expertise.
 - (10) Reduced risk of GEOINT “surprise test findings” at the end.
 - (11) Increased confidence that GEOINT aspects of system development are on track.
- b. Benefits to Program Oversight Bodies. GFMSA assessments provide critical GEOINT-related performance information to senior leadership and oversight bodies, enabling them to manage programs better and arrive at better decisions supported by more detailed GEOINT-specific information. This equates to more informed decisions promoting DoD/IC objectives for cost avoidance and savings. Benefits include:
- (1) Improved insight for the degree to which GEOINT-enabled capabilities are realized in support of military operations.
 - (2) Quantifiable measures for the effective exchange of GEOINT information for both anticipated and unanticipated users.
 - (3) Earlier ability to identify gaps in GEOINT standards conformance to anticipate functional capability gaps likely impacting interoperability.
 - (4) Ease of integration of new GEOINT-aware data source capabilities into the NSG enterprise.
 - (5) Applicability to various acquisition models to include post-acquisition (legacy), incrementally delivered, and fielded systems and system components.

ENCLOSURE 4ASSESSMENT OBJECTIVES AND CRITERIA1. INTRODUCTION

- a. ~~Net-Centric~~ Data and Services Objectives¹ (References ~~c j, m, bb, and e~~)

(1) The ~~general net-centric~~ objectives for any type or variety of data and services are: key enablers necessary for sharing data, information and IT services and ensuring data, information, and IT services are:

- (a) ~~Data/Service is~~ Visible
- (b) ~~Data/Service is~~ Accessible
- (c) ~~Data/Service is~~ Understandable
- (d) ~~Data/Service is~~ Trusted
- (e) ~~Data/Service is~~ Interoperable

(2) GFMSA processes, objectives, and associated criteria specified herein provide the specificity needed to make the above VAUTI objectives GEOINT-aware. These criteria address the actual realization of net-centric data and services objectives within the NSG enterprise enabled from the perspective of GEOINT standards. The realization of these objectives is also dependent on system and infrastructure capabilities enabled by other complementary IT standards needed to task, collect, post, process, manage, protect, serve, exploit, disseminate, discover, access, and use GEOINT.

VAUTI

~~Data, information, and IT services will be made visible, accessible, understandable, trusted, and interoperable throughout their lifecycles for all authorized users.~~

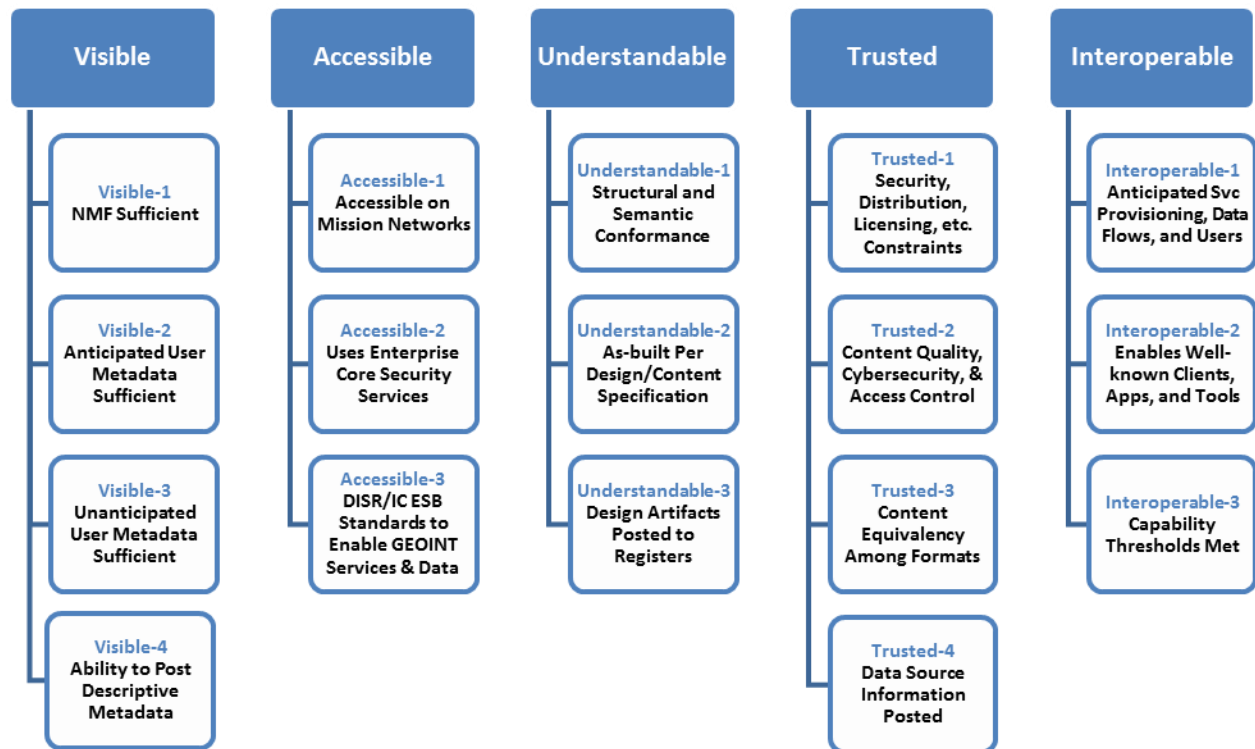
Key enablers necessary for sharing data, information, and IT services and ensuring data, information, and IT services are visible, accessible, understandable, trustworthy, and interoperable. (DoDI 8320.07)

IC elements shall ensure that new IT systems or significant changes to existing IT systems provide the capability for discovery, dissemination, and retrieval of information collected or analysis produced through

¹ ~~DoDI 8320.02, 5 August 2013, Sharing Data, Information and Information Technology (IT) Services in the DoD~~
~~DoD CIO, 4 May 2007, DoD Net-Centric Services Strategy~~
~~DoD CIO Memorandum, 9 May 2003, DoD Net-Centric Data Strategy~~
~~ICD 501, 21 January 2009, Discovery And Dissemination or Retrieval of Information within the Intelligence Community~~
~~IC Information Technology Enterprise Strategy 2012-2017~~

(3) An overview of the data and services objectives and criteria is shown in Figure 3. Each objective has a set of GEOINT-aware criteria for measuring success. These objectives and criteria infuse GEOINT-awareness into the generalized **net-centric** VAUTI objectives for data and services. See Section 4, Objectives, Associated Criteria, and Methods, for detailed descriptions of VAUTI objectives, objective criteria, and assessment confirmation methods.

Figure 3 - Overview of Data/Services Objectives and Criteria



b. Assessing GEOINT Data

(1) Data Assets for Assessment. For GEOINT, the data and metadata assets for assessment consist of one or more instances of each of the following:

(a) GEOINT Data. The data consist of data files, data streams, databases, and other forms for containing GEOINT data. Examples of GEOINT data include still imagery, motion imagery, gridded data, moving features, point cloud data, vector data, feature data, observations, intelligence reports, metadata, etc. The data and metadata is primarily dynamic – variable content specific to the feature, dataset, point, footprint, scene, coverage or instance in time. (Note: The data may also include some static data.)

(b) Static Parameters. A generalized term to describe supplementary parameters and metadata needed to support processing, exploitation, dissemination, and use of the GEOINT data. The data is “static” in that it does not vary on a per-

feature, per-dataset, per-point, per-footprint, per-scene, or per-coverage basis. Static parameters include metadata common to all data collected by a remote sensor system, a specific collection campaign, a specific data series, a specific data collection, a field operation, or similar circumstance for which there is commonality of metadata. Static parameters also include such things as calibration data, common processing parameters and look-up tables, spectro-radiometric parameters, geodetic parameters, code lists/values, and similar data that may not be directly included in the source data.

(c) Data Specification. The data specification describes the content design and population plan for the GEOINT data/metadata and the static parameters specific to the data outputs (e.g., imagery, gridded data, vector data, etc.) of the source system. The specification is static for the assessed configuration.

(d) System Description. The system description provides a general description of the overall system and its GEOINT data-related capabilities. The description is static for the assessed configuration.

Figure 4 depicts the relationships among the data assets to be assessed.

Figure 4. Data Assets Assessed

GEOINT Data (e.g.) • Data File • Data Stream • Database • Metadata • Content is Dynamic	Static Parameters (e.g.) • Metadata • <u>Spectro-Radiometric</u> • Calibration • Look-up Tables / Registers • Code Lists and Values
Data Content Specification • Describes Content Design & Population Plan for: ○ Imagery & Gridded Data ○ Vector Data ○ Other Varieties of Geospatial Data • Content is Static for Assessed Configuration	
System Description • General Description of the System and Its Capabilities • Content is Static for Assessed Configuration	

(2) Data Asset Assessment. There are fundamentally three types of measurements to be performed on the data assets:

(a) Syntax. The data asset is assessed for conformance to the applicable DISR/IC ESB-cited format standards. Data format is essential to the *interpretability* of the data.

(b) Semantics. The data asset is assessed for the *meaningfulness* of data element values. Semantic content is essential to the *understandability* of the data.

(c) Content. The data asset is assessed for sufficiency of information to satisfy the functional capabilities to be enabled by the data. Functional data is essential to the *usability* of the data in a manner responsive to user's needs. Actual performance thresholds (metrics) for assessment are derived from validated/certified system requirements specific to the data assets under assessment.

c. Assessing GEOINT Services

(1) Service Assets for Assessment. For GEOINT web services, service assets for assessment consist of one or more instances of each of the following:

(a) Web Service (Server). The service operation request (command) and response contents, to include parameters, data structures, protocols, and encoding, at the service interface of the implementing system. The content of service operation requests and responses is dynamic and variable according to the operation request or instance in time. Metadata describing the service and its capabilities is comparatively static.

(b) Web Service (Client). The service operation request and response include parameters, data structures and encoding, at the client interface of the implementing system. The content of service operation requests and responses is primarily dynamic—variable content specific to the operation request or instance in time.

(c) Service Design Specification. The service design specification describes the design for provisioning the service and/or client interfaces, their functional capabilities, and the associated metadata of the service providing system. It also addresses rationale for how the design addresses mission effectiveness requirements for the service. The content is static (configuration managed) for the assessed configuration.

(d) System Description. The system description provides a general description of the overall system and its GEOINT-related capabilities. The content is static for the assessed configuration.

Figure 5 depicts the relationships among the service assets to be assessed.

Figure 5. Service Assets Assessed

Web Service (Server) • Service Protocols • Service Commands • Service Responses • Service Metadata • Interface is Dynamic	Web Service (Client) • Service Protocols • Service Commands • Service Responses • Service Metadata • Interface is Dynamic
Service Design Specification • Describes the Service Interface Capabilities and Responses ○ Web Service – Server Side Interface ○ Web Service – Client Side Interface ○ Service Metadata and Content Metadata • Content is Static for Assessed Configuration	
System Description • General Description of the System and Its Capabilities • Content is Static for Assessed Configuration	

(2) Service Asset Assessment. There are fundamentally three types of measurements to be performed on the service assets:

(a) Syntax. The service asset is assessed for conformance to the applicable DISR/IC ESB-cited service operation request/response encoding standards. Service interface encoding is essential to the *interpretability* of the request and responses that characterize the communication between a service and its clients.

(b) Semantics. The service asset is assessed for the meaningfulness of both service and data element values. Semantic content is essential to the understandability of the service metadata. Preserving the data meaningfulness in services is significant because of the potential for the service operation to corrupt the data and destroy semantic meaningfulness.

(c) Content. The service asset is assessed for sufficiency of information content to satisfy the functional capabilities to be enabled by the data. Functional content is essential to the *usability* of the service in a manner responsive to user's needs. Actual performance thresholds (metrics) are derived from validated/certified system requirements specific to the service assets under assessment.

2. TEST AND EVALUATION PLANNING, CONDUCT AND REPORTING

a. GEOINT-Aware T&E Planning. Program and project managers who integrate GEOINT standards conformance assessment into their overall development and testing schedule will experience significantly less risk in successfully transitioning GEOINT-enabled capabilities to those who depend on GEOINT for achieving mission success, and to the NSG community at large. From this perspective, the GFMSA concept of being GEOINT-aware for test planning purposes includes:

(1) Identification and description of the GEOINT-based capabilities that need to be enabled by the information content of the GEOINT data and/or services.

(2) Identification and description of the GEOINT performance requirements, parameters, criteria and metrics that will be used to assess information content sufficiency of the data and/or services.

(3) Identification of the appropriate GEOINT standards needed to enable GEOINT-based capabilities in a manner promoting interoperability objectives across the NSG enterprise.

(4) Identification of specific events needed to measure GFMSA criteria.

(5) Identification of credible test service providers to engage in support of assessment activities and events.

(6) Identification of assessment reporting/submission content sufficient to convey that GFMSA criteria have been successfully met.

b. Acquisition Test Processes. The GFMSA objectives and criteria represent the desired GEOINT-aware end state regardless of the acquisition path or processes used to fund, research, design, develop, implement, test, evaluate, and deploy the GEOINT-enabled component/system. Actual programs/projects need to satisfy various acquisition statutory and regulatory requirements, such as those applicable to Programs of Record (PORs), Quick Reaction Capabilities (QRCs), Joint Capability Technical Demonstrations (JCTDs), etc. Consequently, test planning activities need to be done within the acquisition model pertinent to the specific program/project. The GFMSA approach is designed to work within all existing processes to enable those processes to be GEOINT-aware.

c. Test Planning for Standards Conformance

(1) Conformance with applicable GEOINT standards is an essential activity toward achieving GFMSA objectives. Verifying conformance to GEOINT standards is best accomplished during the system development phase as part of Development Test and Evaluation (DT&E) assessments and operational test readiness reviews.

(2) Program/Project managers conduct assessments and submit evidence substantiating that GFMSA objectives and criteria have been successfully met. Sources of substantiating evidence include results from documentation reviews, analysis, demonstrations, evaluations, examinations, inspections, tests, questionnaires, surveys, conformance registers, collateral sources, and similar activities that confirm conforming implementation of GEOINT standards in a manner enabling interoperability across the NSG enterprise.

(3) There are four types each of data and service assets (described in paragraphs 1.b. and 1.c., respectively) that undergo self-assessment:

Data Assets

- GEOINT Data
- Static Parameters
- Data Content Specification
- System Description

Service Assets

- Web Service (Server)
- Web Service (Client)
- Service Design Specification
- System Description

The criteria and measures for evaluating the data and service assets for information content sufficiency are largely derived from the certified **net-ready performance attribute N-R-KPP** or similar source of authoritative information content requirements.

(4) To assist program/project managers, a number of GEOINT standardization documents have been identified as particularly applicable to GEOINT data and services. Conformance test suites for GEOINT standards exist at different levels of maturity; they vary in their ability to measure all mandatory, conditional and optional provisions of the standards. The state of conformance testability and the associated risk mitigation factors have been evaluated for these standardization documents as an aid to programs/projects in their test planning, conduct and reporting efforts to substantiate standards conformance. Additionally, potential sources for obtaining GEOINT-aware test tools and services have been identified. (See the GFMSA Resource Site).

d. Test Planning for Interoperability. As defined by DoD policy, interoperability is the ability of systems, units, or forces to provide data, information, materiel, and services to, and accept the same from, other systems, units, or forces; and to use the data, information, materiel, and services so exchanged to enable them to operate effectively together. IT and NSS interoperability includes both the technical exchange of information and the end-to-end operational effectiveness of that exchanged information as required for mission accomplishment. Interoperability is more than just information exchange; it includes systems, processes, procedures, organizations, and missions over their life cycle and must be balanced with cybersecurity (formerly Information Assurance).

(1) Interoperability of GEOINT-Related IT

- (a) The methods and processes for asserting conformance to GEOINT standards as enablers of interoperability pertain to:
- i. IT with joint, multinational, and interagency interoperability requirements for which the JITC engages in the process for conducting interoperability test and evaluation; and
 - ii. IT without joint, multinational, and interagency interoperability requirements for which the DoD Components are responsible for conducting interoperability test and evaluation.

(b) The GFMSA strategy is integral to the processes outlined in Reference f p. The GFMSA program seeks to increase GEOINT-awareness in interoperability test and certification activities. It establishes policies and procedures for developing, coordinating, reviewing, and approving IT and NSS interoperability needs. It establishes procedures for performing interoperability certification using a net-ready approach. This manual also promotes the GEOINT FM's objectives for implementation of IC standards needed to enable interoperability across the function's information domains and activities (ICD 113) and furthers IC objectives for discovery and dissemination or retrieval of information within the intelligence community (ICD 501).

(2) Net-Ready ~~Performance Attribute Key-Performance-Parameter~~

(a) The ~~net-ready performance attribute N-R-KPP~~ documents sponsor-identified and JROC-validated verifiable performance measures and metrics for interoperability engineering, design, and testing. To meet ~~net-ready performance attribute N-R-KPP attributes~~, IT must be able to support military operations, to enter and be managed on the network, and to effectively exchange information. The ~~net-ready performance attribute N-R-KPP~~ development process will help verify operationally effective provider-to-consumer, end-to-end information exchanges according to the sponsor's stated capability requirements, applicable reference models, and reference architectures. ~~It informs the solution architecture according to the DoD Information Enterprise Architecture (IEA).~~ (Reference f gg)

(b) Net-ready attributes determine specific measurable and testable criteria for interoperability, and operationally effective end-to-end information exchanges. The ~~net-ready performance attribute N-R-KPP~~ identifies operational, ~~net-centric~~ requirements with threshold and objective values that determine its MOE and MOP. The ~~net-ready performance attribute N-R-KPP~~ covers all communication, computing, and electromagnetic spectrum requirements involving information elements among producer, sender, receiver, and consumer. Information elements include the information, product, and service exchanges. These exchanges enable successful completion of the Warfighter mission or joint business processes. The attributes depict how planned or operational IT:

- i. Attribute 1: Supports military operations.

- ii. Attribute 2: Is entered and managed on the network; and
- iii. Attribute 3: Effectively exchanges information.

(c) The GFMSA process primarily addresses test criteria and measures for the first and third attributes listed above. The criteria are organized according to the DoD Data and Services Strategy objectives of visible, accessible, understandable, trusted and interoperable (VAUTI).

(3) Program of Record. The criteria for GEOINT data and services established herein are pertinent to the ~~net-centric~~ data objectives of the NSG for providers of GEOINT data and/or web services fielded through Program of Record statutory and regulatory information requirements, acquisition and test process procedures.

(4) Quick Reaction Capability (QRC). DoD acquisition directives and instructions authorize Milestone Decision Authorities to tailor (streamline) the regulatory information requirements, acquisition and test process procedures to achieve cost, schedule, and performance goals. The GFMSA criteria established herein remain pertinent to the ~~net-centric~~ data and services objectives of the NSG for providers of GEOINT data and/or web services fielded through QRC processes.

(5) Joint Capability Technology Demonstration (JCTD). The criteria identified herein are pertinent to the GEOINT aspects of test planning for JCTD activities.

(6) Multinational and Coalition. Test planning to address compliance with international and NSG-adopted standards and interoperability criteria is essential to achieve GEOINT IT standards-enabled interoperability with the entirety of the NSG in accordance with in-force international treaties, Memorandums of Agreement (MOAs)/Memorandums of Understanding (MOUs), and national laws, orders and directives.

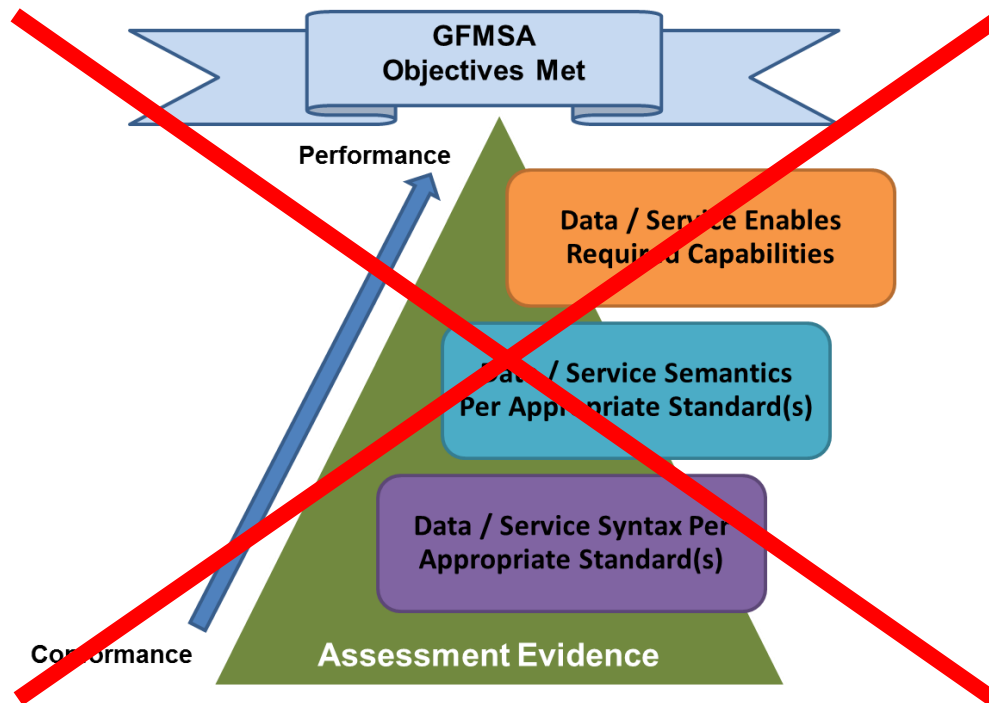
3. DEVELOPMENTAL AND OPERATIONAL TEST AND EVALUATION

a. Integration Expectations. As stated previously, the GFMSA approach is designed to work within existing processes to enable those processes to be GEOINT-aware. Integrating GFMSA objectives and criteria into T&E strategies and master plans enables Developmental Test and Evaluation (DT&E) and Operational Test and Evaluation (OT&E) testing and test results to support GEOINT-aware joint interoperability test certification and eliminate duplication. Program and project managers coordinate with NGA on all GEOINT-related requirements, ISPs, test strategies and plans, test and evaluation results, and interoperability certifications. (Reference ~~d k~~).

b. T&E Conduct. Test and Evaluation events used to assess GFMSA objectives and criteria will often span DT&E, OT&E, interoperability, and other test activities, relying on multiple test events conducted by various test organizations. The amount and type of testing will vary based on characteristics of the component/system being evaluated.

Developmental testing performed under government supervision that generates reliable, valid data can be used to determine technical capabilities and standards conformance status, and may supplement operational data for a GEOINT-aware interoperability evaluation. Each potential data collection opportunity should be used in the overall T&E process to get the best GEOINT-aware, performance-based picture of the component/system in the most efficient manner possible to substantiate GFMSA objectives and criteria. ~~as notionally depicted in Figure 6.~~

~~Figure 6—Testing for Performance~~



c. T&E Reporting. Throughout the acquisition cycle, evidence from developmental test, operational test and evaluation, documentation reviews, demonstrations, evaluations, examinations, field exercises, questionnaires, surveys, and/or other reliable sources may be used to substantiate that GFMSA objectives and criteria are in the process of being met, or have been successfully met (GEOINT-aware reporting). Complex systems involving multiple evaluation events may benefit from preparation of a capstone report which outlines how the component/system test results substantiate meeting GFMSA objectives.

4. OBJECTIVES, ASSOCIATED CRITERIA, AND METHODS

a. The following Tables articulate the GFMSA objectives, associated criteria, and methods used to confirm success in meeting each GFMSA criteria. In some cases, alternative means for meeting the GFMSA criteria are identified. The methods for meeting criteria may need to be tailored for specific applications. In such cases, an additional means for meeting GFMSA criteria may be nominated to the GFMSA Manager(s) prior to conduct of test events.

b. Data/Service is Visible. Objective: Users, applications and services can discover the existence of GEOINT data and web service assets through catalogs, registries, and other search services. GEOINT data and service assets are advertised, or “made visible”, by providing metadata, which describes the asset. See Table 3.

Table 3. Data/Service is Visible	
Objective Criteria	Confirmation Methods
<u>Visible-1</u> NMF Sufficient <i>Data:</i> Data element content of the data asset (dynamic) and associated description artifacts (static) is sufficient to permit creation of an NSG Metadata Foundation (NMF)-conforming description (e.g., metacard) of the data asset. <i>Service:</i> An NMF-conforming description of the service asset is published to one or more enterprise service-discovery registry.	Confirmed by all of the following: - Data element content conforms to the NMF-conforming encoding. (NGA.STND.0012 series of conceptual metadata standards). - Data element content satisfies the criteria found in the NMF Abstract Test Suites (ATs) of the NGA.STND.0012 series of standards and NMF-conforming encoding specification(s) ATs. - Metadata extensions, if any, conform to the ISO 19115 rules for metadata extensions, which the NMF extension strategy inherits from ISO. - Service description metadata is made available via the provisioned service interface. - For services providing access to GEOINT data, data description metadata is made available via the provisioned service interface. - Information content is conceptually equivalent to its NMF counterpart, regardless of encoding. Note: Prominent DISR/IC ESB-adopted encodings of NMF-conforming data and/or service asset descriptions include: NMIS, DDMS, and TDF.
<u>Visible-2</u> Anticipated User Metadata Sufficient. <i>Data:</i> Data element content of the data asset (dynamic) and associated description artifacts (static) includes additional discovery and descriptive metadata elements beyond the NMF minimum set pertinent to anticipated users and uses of the data asset. <i>Service:</i> Data element content of the service asset description includes additional discovery and description metadata elements beyond the NMF minimum set pertinent to anticipated users and uses of the service asset.	Confirmed by all of the following: - Anticipated users are identified in program requirements, acquisition, and/or developmental documentation. - Sufficiency of discoverability by anticipated users assessed and reported by validation test events that include anticipated users. - Applicable metadata are first selected from the NMF-established vocabulary (NSG.STND.0012 series of standards); and then from the ISO 19115 vocabulary as available. - Metadata extensions, if any, conform to the ISO 19115 rules for metadata extensions, which the NMF extension strategy inherits from ISO.

Table 3. Data/Service is Visible	
Objective Criteria	Confirmation Methods
<u>Visible-3</u> Unanticipated User Metadata Sufficient. <i>Data:</i> Information within the data and associated metadata artifacts includes data lineage, data quality, and data accuracy elements to assist usability assessment of the data for a given purpose (unanticipated users). <i>Service:</i> Service interfaces provide self-describing metadata (e.g., GetCapabilities response) to include metadata description of available data assets and/or processing services (unanticipated users).	Confirmed by all of the following: - Data element content conforms to minimal content requirements of NGA.STND.0012-1, NSG Metadata Foundation – Part 1: Core, and relevant parts in the NMF series of standards. - Data element content conforms to optional lineage, quality, and accuracy content requirements of NGA.STND.0012-1, NSG Metadata Foundation – Part 1: Core and the relevant parts in the NMF series of standards. - Data element content conforms to extended content requirements of NGA.STND.0012-1, NSG Metadata Foundation – Part 1: Core and the relevant parts in the NMF series of standards.
<u>Visible-4</u> Ability to Post Descriptive Metadata. <i>Data/Service:</i> Demonstrated ability to post discovery metadata about data and/or service assets into one or more catalogs, registries, repositories, or other search services within the NSG enterprise.	Discovery metadata submission package successfully input into one or more of the following repositories: - National Information Library (NIL) - Command Information Library (CIL) - St. Louis Information Library (STIL) - DCGS Integrated Backbone (DIB) Metadata Catalog - Coalition Shared Data (CSD) Server - Consolidated Analytic Spatial-Data Initiative (CASI) - NSG Consolidated Library (NCL) - Library of National Intelligence (LNI) - Consolidated Production Database (CPD) - Others (Identify)

c. Data/Service is Accessible

(1) Objective: Users, applications, and other services ~~have the~~ ability to access GEOINT services and data in a timely, secure, and effective manner. Accessibility implies that:

(a) Descriptive information about the data and/or service asset (metadata) has been provided to a catalog that is visible to the Enterprise; and

(b) The data and/or services are provisioned such that users, applications, and other services in the Enterprise can access it.

(2) Service and data assets are made available to any user, application, or other service except when limited by policy, regulation, or security. See Table 4.

Table 4 – Data/Service is Accessible	
Objective Criteria	Confirmation Methods

Table 4 – Data/Service is Accessible	
Objective Criteria	Confirmation Methods
<u>Accessible-1</u> Accessible on Mission Networks <i>Data/Service:</i> Demonstrated ability to post, access, and use the GEOINT service and/or data assets on the mission-required networks.	Confirmed by one or more of the following: - Receipt of Authority to Connect (ATC). - Receipt of Interim Authority to Connect (IATC). - DT&E test event report. - OT&E test event report. - Interoperability Certification test report. - Field exercise or demonstration report. - Test facility replication of fielded systems and interfaces.
<u>Accessible-2</u> Uses Enterprise Core Security Services. <i>Data/Service:</i> Provider uses the core security services provided by the Enterprise Information Environment Mission Area (EIEMA) to ensure that widest possible access is supported to authorized users.	Confirmed by all of the following: - Design verification that the service is qualified to operate on the given network using EIEMA-provisioned security services. - When implementing XACML as the access control policy language, GeoXACML GML Extensions are used for geospatial-related access control. - Verification that users with proper credentials and authentication are able to access services and data. - Verification that users without proper credentials and authentication are not able to access neither services nor data.
<u>Accessible-3</u> DISRIC/IC ESB Standards to Enable GEOINT Services and Data. <i>Data/Service:</i> Service uses open, non-proprietary DISR/IC ESB-listed standards (e.g., World Wide Web Consortium (W3C), Organization for the Advancement of Structured Information Standards (OASIS), Open Geospatial Consortium (OGC) Web standards) to enable access to GEOINT services and data for consumers.	Confirmed by all of the following: - Design verification that service uses DISR/IC ESB-cited standards. - HTTP-based protocols (e.g., REST XML, REST Keyword Value Pair (KVP), SOAP XML, etc.) are used for exchanging structured information as called for in the GEOINT standards applicable to the service network interface. - WSDL is used to automate the details involved in applications communication as called for in the GEOINT standards applicable to the service network interface. - The service is discoverable on a UDDI-based web services registry pertinent to the network environment on which the service operates. - Detailed geospatial queries are provided using OGC CSW details in the OGC Catalog Services – Web standard.

d. Data/Service is Understandable. Objective: Users, applications, and other services can comprehend the data and/or service interface protocol, both structurally and semantically, and readily determine how the data and/or service may be used for their specific needs. See Table 5.

Table 5 – Data/Service is Understandable	
Objective Criteria	Confirmation Methods

Table 5 – Data/Service is Understandable	
Objective Criteria	Confirmation Methods
<u>Understandable-1</u> Structural and Semantic Conformance. <i>Data:</i> The data is structurally and semantically compliant with the pertinent GEOINT data formatting and content standards cited in the DISR/IC ESB. <i>Service:</i> The service interface protocol and associated schema are structurally and semantically conformant with the pertinent GEOINT service standards cited in the DISR/IC ESB.	Confirmed by all of the following: - Basic format (syntax and semantics) standards conformance assessment successfully completed. - Capability-specific implementation profile conformance assessment successfully completed (e.g., EO, SAR, LiDAR, Terrain Model, NAS, Web Service Profile, etc.).
<u>Understandable-2</u> As-built Per Design/Content Specification. <i>Data:</i> The as-built for deployment data content is in full alignment with its Data Content Specification. <i>Service:</i> The as-built for deployment GEOINT service is in full alignment with its Service Design Specification to include the vocabularies and business rules that underlie the implementation of the service.	Confirmed by all of the following: - For all modes of sensor operation, data production, and service provisioning options, test results of post-IOC, operationally-produced data correspond with test results of pre-IOC data outputs (which may have been simulated). - The Data Content Specifications and/or Service Design Specifications accurately reflect the operational data and/or service assets.
<u>Understandable-3</u> Design Artifacts Posted to Registers. <i>Data/Service:</i> The design artifacts are structurally and semantically documented in a form acceptable to the maintenance authority for the DoD and/or IC registers to which they are posted. Design artifacts include such as: - System Description - Data Content Specification - Service Design Specification - Documentation of data structures, models, dictionaries, schemas, parameters and similar information.	Confirmed by all of the following: - The System Description is posted on the DoD DSE, the IC ESB ER2, and/or another authorized, operationally-accessible register. (Identify the registers). - The Data Content Specifications and/or Service Design Specifications are posted on the DoD DSE, the IC ESB ER2, and/or another authorized, operationally-accessible register. (Identify the registers). - Documentation of data structures, models, dictionaries, schemas, parameters and similar are posted on the DoD DSE, the IC ESB ER2, and/or another authorized, operationally-accessible register. (Identify the registers). - The means to make available operationally-needed schemas, parameters, and similar artifacts to operations without access to network services has been established.

e. Data/Service is Trusted. Objective: Users, applications, and other services can determine and assess the authority of GEOINT data and services because the pedigree, security level, and access control level of each data and service asset is known and available. See Table 6.

Table 6 – Data/Service is Trusted	
Objective Criteria	Confirmation Methods

Table 6 – Data/Service is Trusted

<u>Trusted-1</u> Security, Distribution, Licensing, etc. constraints. <i>Data:</i> Security, distribution, licensing, and other constraint marking content for data assets complies with the classification guide applicable to the data source as established by security classification professionals. <i>Service:</i> Service descriptive metadata identifies the security, distribution, licensing, and other constraints pertinent to the proffered services and associated data as established by security classification professionals.	Confirmed by all of the following: - The System Description accurately identifies the classification guide applicable to the data and/or service assets implemented by the system. - The Data Content Specifications and/or Service Design Specifications identify all applicable security and distribution data elements pertaining to the output data and/or service, and the Controlled Vocabulary Enumerations (CVE) pertinent to the data elements. - A review case has been completed by a security classification management specialist attesting to the completeness and sufficiency of the specified security and distribution marking content for the data and/or service assets.
<u>Trusted-2</u> Content Quality, Cybersecurity, and Access Control. <i>Data:</i> Quality information for geospatial content is expressly included in the data and/or its associated data artifacts. <i>Service:</i> The services provisioning system is accredited for cybersecurity and secure operations, to include role and/or attribute-based access control (RBAC/ABAC) processes on deployed networks.	Data criteria confirmed by one or more of the following: - Accuracy (degree of confidence) information for geospatial content is expressly included in the data and/or its associated data artifacts. - For sensor data, sensor geometry and error propagation model is documented and reviewed for sufficiency against certified performance requirements by an accredited photogrammetric scientist. - Quality parameters needed for geospatial processing are present. - Operationally realized accuracy confirmed against geospatial truth. - The System Description documents realized accuracy characteristics. Service criteria confirmed by: - The service provider's computing system is accredited for operation on the network (or stand-alone operation when operationally needed). - Controlled Unclassified Information (CUI) and classified information is accessible only by granting access to persons and/or non-person entities through RBAC and/or ABAC processes.
<u>Trusted-3</u> Content Equivalency Among Formats. <i>Data/Service:</i> For data sources and/or services opting to provide data in more than one DISR/IC ESB-cited format, a Data Content Specification is registered describing each formatting option, and the System Description addresses the degree of information content equivalency among the supported formats.	Confirmed by all of the following: - The Data Content Specifications describe the degree of information content equivalency among the supported formats. - Each Data Content Specification is duly registered. - Metadata description of the data series summarizes significant differences of information content, if any, among the supported formats.
<u>Trusted-4</u> Data Source Information is Posted. <i>Data/Service:</i> Data source information is entered into the DoD and/or IC registers of known/authorized data sources.	Confirmed by one or more of the following: - The System Description is posted on the DoD DSE. - The System Description is posted on the IC ESB ER2. - The System Description is posted on another authorized operationally accessible register. (Identify the register.)

f. Data/Service is Interoperable. Objective: GEOINT services and data are accessed by users, applications, or other services that are sometimes predefined or sometimes unanticipated. Metadata and protocols are available to enable negotiation of service capability options, common information exchange reference models, and mediation or translation of data between GEOINT-standards conforming interfaces (e.g., server and client), as needed. See Table 7.

Table 7 – Data/Service is Interoperable	
Objective Criteria	Confirmation Methods
<u>Interoperable-1</u> Anticipated Service Provisioning, Data Flows, and Users. <i>Data:</i> Demonstrated ability to exchange data among elements and across interfaces identified in acquisition and associated requirements documentation (anticipated data flows and users). <i>Service:</i> Demonstrated ability to provide the services, make use of DoD/IC-mandated Core Enterprise Services (CES) and other applicable enterprise services, and exchange associated data among elements and across interfaces identified in acquisition and associated requirements documentation (anticipated service provisioning, data flows and users).	Confirmed by one or more of the following: a. OT&E test event report. b. Interoperability Certification test report. c. Field exercises or demonstrations. d. Test facility replication of fielded systems and interfaces.
<u>Interoperable-2</u> Enables Well-known Clients, Applications, and Tools. <i>Data/Service:</i> Demonstrated ability to enable basic service access and data content visualization by common, well-known service clients, user applications and tools (unanticipated users).	Confirmed by one or more of the following: a. Standards conformance test event report. b. DT&E test event report. c. OT&E test event report. d. Interoperability Certification test report. e. Field exercises or demonstrations report. f. Test facility replication of fielded systems and interfaces.
<u>Interoperable-3</u> Capability Thresholds Met. <i>Data/Service:</i> Demonstrated ability to meet capability, effectiveness and performance thresholds as stated in approved acquisition requirements documentation (anticipated users).	Confirmed by one or more of the following: a. Standards conformance test event report. b. DT&E test event report. c. OT&E test event report. d. Joint Interoperability Certification test report. e. Field exercises or demonstrations report. f. Test facility replication of fielded systems and interfaces.

GLOSSARYPART I. ACRONYMS~~1O1T — One Object One Time~~

ABAC Attribute Based Access Control

~~ABI — Activity Based Intelligence~~~~ACAT — Acquisition Category~~~~ADS — Authoritative Data Source~~

AOR Area of Responsibility

~~ASG — Allied System for Geospatial Intelligence~~

ATC Authority to Connect

ATS Abstract Test Suite

AT&L Acquisition Technology and Logistics

~~AV — All View~~~~C2 — Command and Control~~~~C4 — Command, Control, Communications, and Computers~~~~CA — Capability Architecture~~

CAC Common Access Card

CASI Consolidated Analytic Spatial-Data Initiative

CCIF Combatant Command Initiatives Fund

CCMD Combatant Commands

~~C/S/A — Combatant Commands, Services, Agencies~~

CDD Capability Development Document

CES Core Enterprise Services

CIL Command Information Library

CIO Chief Information Officer

CJCS Chairman of the Joint Chiefs of Staff

CJCSI Chairman of the Joint Chiefs of Staff Instruction

~~CJCSM — Chairman of the Joint Chiefs of Staff Manual~~

CM Configuration Management

~~COCOM — Combatant Command~~~~COI — Community of Interest~~

CONOPS Concept of Operations

COTS Commercial–Off-the-Shelf

CPD Capabilities Production Document

CPD Consolidated Production Database

~~CPM — Capability Portfolio Manager/Management~~

CSA Combat Support Agency

CSD Coalition Shared Data (Server)

CSW Catalog Services Web

CUI Controlled Unclassified Information

CVE Controlled Vocabulary Enumerations

DAG	DoD Acquisition Guidebook
DAU	DoD Acquisition University
DBS	Defense Business System
DCGS	Distributed Common Ground System
DCI	Director of Central Intelligence
DCR	DOTmLPF-P (Doctrine, Organization, Training, materiel, Leadership and Education, Personnel, Facilities-Policy) Change Recommendation DOTMLPF Change Recommendations
DDMS	Department of Defense Discovery Metadata Specification
DHS	Department of Homeland Security
DI2E	Department of Defense Intelligence Information Environment
DIA	Defense Intelligence Agency
DIB	DCGS Integrated Backbone
DISA	Defense Information System Agency
DISR	DoD Information Technology Standards Registry
DoD	Department of Defense
DoD CIO	Department of Defense Chief Information Officer
DoD IEA	Defense Information Enterprise Architecture
DoDAF	DoD Architecture Framework
DoDD	DoD Directive
DoDI	DoD Instruction
DOTmLPF-P	Doctrine, Organization, Training, materiel, Leadership and Education, Personnel, Facilities-Policy
DOTMLPF	Doctrine, Organization, Training, Materiel, Leadership and Education, Personnel, and Facilities
DNI	Director of National Intelligence
DSE	(DoD) Data Services Environment
DT	Developmental Testing
DT&E	Developmental Test and Evaluation
DTM	Directive-Type Memorandum
D/NGA	Director, National Geospatial-Intelligence Agency
EIE	Enterprise Information Environment
EIEMA	Enterprise Information Environment Mission Area
EISP	Enhanced Information Support Plan
EO	Electro-Optical
ER2	(IC) Enterprise Registry and Repository
ESB	(IC) Enterprise Standards Baseline
FM	Functional Manager
FY	Fiscal Year
FYDP	Fiscal Year Defense Program
GEOCOM	National Geospatial Intelligence Committee
GEOINT	Geospatial Intelligence
GEOINT FM	GEOINT Functional Manager

GFMSA	GEOINT Functional Manager Standards Assessment
GIG	Global Information Grid
GML	Geography Markup Language
GOTS	Government Off The Shelf (Software)
GSD	Ground Sample Distance
GTG	GIG Technical Guidance
GTG-F	GTG Federation
GWG	Geospatial Intelligence Standards Working Group
HTTP	Hyper Text Transfer Protocol
IA	Information Assurance
IACT	Interim Authority to Connect
IAW	In Accordance With
IC	Intelligence Community
ICAM	Intelligence Community Acquisition Model
ICCR	Intelligence Community Capability Requirements
ICD	Initial Capabilities Document
IC ITE	Intelligence Community Information Technology Enterprise
ICPG	Intelligence Community Policy Guidance
IEA	Information Enterprise Architecture (DoD)
IOC	Initial Operational Capability
IOP	Interoperability
IP	Internet Protocol
IPL	Image Product Library
IRC	Intelligence Requirements Certification
IS	Information System
ISO	International Organization for Standardization
ISP	Information Support Plan
ISR	Intelligence, Surveillance and Reconnaissance
IT	Information Technology
ITE	(IC) Information Technology Enterprise
JCA	Joint Capability Area
JCS	Joint Chiefs of Staff
JCIDS	Joint Capabilities Integration and Development System
JCTD	Joint Capability Technology Demonstration
JESC	Joint Enterprise Standards Committee
JIE	(DoD) Joint Information Enterprise
JIM	Joint, Interagency, Intergovernmental and Multi-national
JITC	Joint Interoperability Test Command
JROC	Joint Requirements Oversight Council
KPP	Key Performance Parameter
KVP	Key Value Pair

LiDAR	Light Detection and Ranging
LNI	Library of National Intelligence
M&S	Modelling & Simulation
MDA	Milestone Decision Authority
MOE	Measure of Effectiveness
MOP	Measure of Performance
MS	Milestone
NATO	North Atlantic Treaty Organization
NCL	NSG Consolidated Library
NGA	National Geospatial-Intelligence Agency
NIIRS	National Imagery Interpretability Rating Scale
	NATO Imagery Interpretability Rating Scale
NIL	National Information Library
NIPRNET	Non-secure Internet Protocol Router Network
NMF	NSG Metadata Foundation
NMIS	NSG Metadata Implementation Specification
N-R KPP	Net Ready Key Performance Parameter
NSG	National System for Geospatial Intelligence
NSGD	National System for Geospatial Intelligence Directive
NSGM	National System for Geospatial Intelligence Manual
NSMC	NSG Senior Management Council
NSS	National Security Systems
OASIS	Organization for the Advancement of Structured Information Standards
OBP	Object Based Production
OGC	Open Geospatial Consortium
OSD	Office of the Secretary of Defense
OT	Operational Test
OTA	Operational Test Agency
OT&E	Operational Test and Evaluation
PDF	Portable Document Format
PED	Processing, Exploitation, and Dissemination
PIV	Personal Identity Verification (certificate)
PM	Program Manager
POC	Point Of Contact
POR	Program of Record
QRC	Quick Reaction Capability
RBAC	Role Based Access Control
REST	Representational State Transfer
SAP	Special Access Program

SAR	Side-looking Airborne Radar
_____	Synthetic Aperture Radar
SecDef	Secretary of Defense
SIPRNET	SECRET Internet Protocol Router Network
SOC	Statement of Capability
SOW	Statement of Work
SPS	Sensor Planning Service
StdV	Standards View
STND	Standard
STIL	St. Louis Information Library
TBD	To Be Determined
TDF	Trusted Data Format
T&E	Test and Evaluation
TEMG	Test and Evaluation Management Guide
TEMP	Test and Evaluation Master Plan
TES	Test and Evaluation Strategy
TV	Technical Standards View
UDDI	Universal Description Discovery Integration
USD (AT&L)	Under Secretary of Defense (Acquisition, Technology, and Logistics)
USD (I)	Under Secretary of Defense (Intelligence)
VAUTI	Visible, Accessible, Understandable, Trusted and Interoperable
W3C	World Wide Web Consortium
WSDL	Web Services Description Language
WWW	World Wide Web
XACML	eXtensible Access Control Markup Language
XML	eXtensible Markup Language
XSD	XML Schema Definitions

PART II. DEFINITIONS

Abstract Test: A test stated in human-understandable terms without reference to a specific item, implementation, or system under test. In contrast, an executable test is derived from an abstract test. It includes lower-level, concrete details needed to interact with and test a specific item, implementation or system under test within its implemented environment.

Abstract Test Suite (ATS): The set of abstract tests covering all specific requirements to be satisfied for the conformance of an implementation under test (ISO19105). An

ATS that addresses NSG-specific requirements is categorized as “NSG Sufficient”; in some cases the available ATS supports only “General Purpose” application for the related standards.

Analysis: The use of recognized analytic techniques (including computer models) to interpret or explain the behavior/performance of the system element. Analysis of test data, or review and analysis of design data, is used, as appropriate, to verify requirements. (Reference I) (DAG)

Architecture: The organizational structure and associated behavior of a system. Architecture can be recursively decomposed into parts that interact through interfaces, relationships that connect parts, and constraints for assembling parts. Parts that interact through interfaces include classes, components, and subsystems.

Assessment: The systematic process of obtaining and evaluating evidence from documentation reviews, analysis, demonstrations, evaluations, examinations, tests, questionnaires, surveys, conformance registers, and collateral sources to make a substantiated assertion about the progress made by a system (or its elements) development activity to conformingly implement GEOINT standards in a manner enabling interoperability across the NSG enterprise.

~~**Authoritative Data Source (ADS):** Those sources registered in the DoD Data Services Environment (DSE) (accessible at <https://metadata.ces.mil/dse>). (DoDI 8320.02). The DSE defines authorizing bodies: “An Authoritative Body that has the Data Need, and is the one that has the authority to declare a given system authoritative for the Data Need. COCOMS, Services, Agencies, CPMs and COIs can be Authoritative bodies.”~~

~~**Capability Architecture:** A set of descriptions that portrays the context and rules required to achieve a desired effect through a combination of doctrine, organization, training, materiel, leadership and education, personnel, and facilities. (Reference gg) (DoDAF V2.0)~~

~~**Capability:** The ability to achieve a desired effect under specified standards and conditions through combinations of means and ways across the doctrine, organization, training, materiel, leadership and education, personnel, and facilities (DOTMLPF) to perform a set of tasks to execute a specified course of action. It is defined by an operational user and expressed in broad operational terms in the format of an Initial Capabilities Document or a joint DOTMLPF change recommendation. In the case of materiel proposals/documents, the definition will progressively evolve to DOTMLPF performance attributes identified in the Capability Development Document and the Capability Production Document. (Reference f) (CJCSI 3170.01)~~

~~**Cybersecurity:** Prevention of damage to, protection of, and restoration of computers, electronic communications systems, electronic communications services, wire communication, and electronic communication, including information contained therein, to ensure its availability, integrity, authentication, confidentiality, and nonrepudiation.~~

~~(DoD Instruction 8500.01, Cybersecurity as quoted from National Security Presidential Directive-54/Homeland Security Presidential Directive-23)~~

Defense Agencies: All agencies and offices of the Department of Defense, including the Missile Defense Agency, Defense Advanced Research Projects Agency, Defense Commissary Agency, Defense Contract Audit Agency, Defense Finance and Accounting Service, Defense Information Systems Agency, Defense Intelligence Agency, Defense Legal Services Agency, Defense Logistics Agency, Defense Threat Reduction Agency, Defense Security Cooperation Agency, Defense Security Service, National-Geospatial Intelligence Agency, National Reconnaissance Office, and National Security Agency/Central Security Service.

Demonstration: The performance of operations at the system, or system element, level where visual observations are the primary means of verification. Demonstration is used when quantitative assurance is not required for verification of the requirements.

~~(Reference I) (DAG)~~

DoD Components: A collective reference to: the Office of the Secretary of Defense (OSD), the Military Departments, the Office of the Chairman of the Joint Chiefs of Staff (CJCS) and the Joint Staff, the Combatant Commands (CCMDs), the Office of the Inspector General of the Department of Defense, the Defense Agencies, the DoD Field Activities, and all other organizational entities within the DoD (referred to in this instruction as the "DoD Components"). ~~(Reference d) (DoDI-8330.01)~~

DoD Enterprise: Relating to policy, guidance, or other overarching leadership provided by OSD Officials and the Chairman of the Joint Chiefs of Staff in exercising authority, direction, and control of their respective elements of the Department of Defense on behalf of the Secretary of Defense.

DoD GEOINT Manager: D/NGA's role as prescribed by DoDD 5105.60 to conduct all GEOINT activities delegated under the authorities of the SecDef ~~EF~~ and Undersecretary of Defense for Intelligence [USD(I)] in accordance with NGA's statutory missions under Titles 10 and 50 United States Code. ~~(Reference b) (DoDD-5105.60)~~

DoD Information Enterprise Architecture: A federation of descriptions that provide context and rules for accomplishing the mission of the Department. These descriptions are developed and maintained at the Department, Capability Area, and Component levels and collectively define: (a) the people, processes, and technology required in the "current" and "target" environments, and (b) the roadmap for transition to the target environment.

DoD Information Technology Standards Registry (DISR): A registry of IT standards which are selected through a defined governance process. It contains the minimal set of rules governing the arrangement, interaction, and interdependence of IT system parts or elements, whose purpose is to ensure that a conformant system satisfies a specified set of requirements. It defines the service areas, interfaces, standards (DISR elements), and standards profiles applicable to all DoD systems. Use of the DISR is

mandated for the development and acquisition of new or modified fielded IT systems throughout the DoD. The standards cited in the DISR replaced the Joint Technical Architecture. (Reference d) ~~(DoDI 8330.01)~~

~~**Enhanced Information Support Plan (EISP):** Use of the EISP is encouraged to facilitate the development of standard ISP formats and assist programs in risk mitigation. The EISP tool is a desktop software application that provides a standard methodology for discovery, analysis, and management of an acquisition program's information dependencies. Data entered into the EISP tool will be tagged with XML. The tagging is transparent to the user and requires no PM's actions but enables the data to be easily stored, searched, retrieved, and reused. The EISP process uses a predefined output script that automatically creates a PDF ISP document. Information on the EISP tool is available on the CJCSI 6212 Resource Page. (Appendix E to Enclosure D of the JCIDS Manual, 12 February 2015, "Content Guide for the Net-Ready Key Performance Parameter (N-R KPP)")~~

Enterprise Architecture: A strategic information asset base that defines the mission, technology, and information necessary to perform the mission and a transitional process for implementing new capabilities in response to changing needs.

Enterprise Information Environment (EIE): The common, integrated information computing and communications environment of the GIG. The EIE is composed of GIG assets that operate as, provide transport for, and/or assure local area networks, campus area networks, tactical operational and strategic networks, metropolitan area networks, and wide area networks. The EIE includes computing infrastructure for the automatic acquisition, storage, manipulation, management, control, and display of data or information, with a primary emphasis on DoD enterprise hardware, software operating systems, and hardware/software support that enable the GIG enterprise. The EIE also includes a common set of enterprise services, called Core Enterprise Services, which provide awareness of, access to, and delivery of information on the GIG. (Reference s) ~~(DoDD 8115.01)~~

~~**Enterprise Registry and Repository (ER2):** Supports the governance of the IC Enterprise Standards Baseline (ESB), and the publication, change, discovery, and retrieval of the IC ESB within the IC Enterprise. These assets provide prescriptive guidance to program and developer for promoting interoperability, information sharing and reuse.~~

Enterprise Standards Baseline (ESB): The DoD and IC Joint Enterprise Standards Baseline (ESB) is the formally identified minimal set of enterprise standards determined to align with and facilitate implementation of an organization's enterprise architecture. Compliance with the organization's enterprise architecture is measured, in part, by adherence of information resources and enterprise architecture-related IT items to relevant standards in the enterprise standards baseline. (Reference r) ~~(ICSI 500-20)~~

Evaluation: Denotes the process whereby data are logically assembled, analyzed, and compared to expected performance to aid in decision making. It may involve review

and analysis of qualitative and quantitative data obtained from design reviews, hardware inspections, modeling and simulation (M&S), hardware and software testing, metrics review, and operational usage of equipment. (Reference m) ~~(DAU Glossary)~~

Examination: Visual inspection of equipment and evaluation of drawings and other pertinent design data and processes used to verify conformance with characteristics such as physical, material, part and product marking, and workmanship. (Reference l) ~~(DAG)~~

Fielded System: A post-acquisition IT/NSS in use by operational or headquarters units (regardless of the process used to put it into operational use). Fielded systems may be modified or improved through standard DoD and/or IC processes.

Functional Manager (FM): FMs are the principal advisors to DNI on the performance of their respective functions, and they serve as the cognizant authorities with respect to the overall performance of their functions within and across IC elements and activities. (Reference a) ~~-as further described in ICD 113-~~

GEOINT Capabilities: The three elements of GEOINT (imagery, imagery intelligence, and geospatial information) that provide a strong foundation for GEOINT capabilities. The combination of the three elements creates a synergy which allows enhanced GEOINT capabilities to:

- a. Use Advanced Sensors
- b. Employ Multiple Sensors Simultaneously
- c. Use Multiple INTs
- d. Expand Visualization Features
- e. Enhance Precision and Detail
- f. Improve On-Demand Global Access (Reference k) ~~(NSG Pub 1-0)~~

GEOINT Operations: Tasks, activities, and events necessary to collect, manage, analyze, generate, visualize, and provide imagery, imagery intelligence, and geospatial information necessary to support national and defense missions and international arrangements.

GEOINT Standard: A documented agreement containing technical specifications or other precise criteria to be used consistently as rules, guidelines, or definitions of characteristics to ensure that materials, products, processes, or services are fit for the analysis and visual representation of physical features and geographically Referenced activities.

Geospatial Information: Information that identifies the geographic location and characteristics of natural or constructed features and boundaries on the earth and includes: statistical data and information derived from, among other things, remote

sensing, mapping, and survey technologies; and mapping, charting, geodetic data and related products. (Reference q)

Geospatial Intelligence (GEOINT): The exploitation and analysis of imagery and geospatial information to describe, assess, and visually depict physical features and geographically referenced activities on the earth. ~~Geospatial intelligence GEOINT~~ consists of imagery, imagery intelligence, and geospatial information. (Reference q) ~~(Title 10 U.S. Code §467)~~

Geospatial Intelligence (GEOINT) Functional Manager: Refers to the D/NGA role as the manager for U.S. GEOINT interests as prescribed by Executive Order 12333 and further delegated by the DNI (Reference a) and SecDef (Reference b) ~~in ICD 113 and DoDD 5105.60.~~

Geospatial Intelligence Functional Manager Standards Assessment (GFMSA): ~~GFMSA defines and implements NSG methods and processes to assess and assert Information Technology (IT) and National Security Systems (NSS), referred to as IT, conformance with GEOINT data and service standards within the Department of Defense (DoD) and/or the Intelligence Community (IC) (Reference j) The GFMSA procedures define NSG methods and processes for asserting IT conformance to GEOINT standards as key enablers of interoperability across the NSG enterprise. Successful completion of the GFMSA process results in community-wide acknowledgement that an IT component has been assessed in a credible manner and found to meet the standards conformance and interoperability criteria set forth by the NSG community, thus optimizing the potential for a system to meet its GEOINT-related operational performance objectives.~~

GEOINT Capabilities: ~~The three elements of GEOINT (imagery, imagery intelligence, and geospatial information) that provide a strong foundation for GEOINT capabilities. The combination of the three elements creates a synergy which allows enhanced GEOINT capabilities to:~~

- ~~g. Use Advanced Sensors~~
- ~~h. Employ Multiple Sensors Simultaneously~~
- ~~i. Use Multiple INTs~~
- ~~j. Expand Visualization Features~~
- ~~k. Enhance Precision and Detail~~
- ~~l. Improve On-Demand Global Access (Reference v) (NSG Pub 1-0)~~

GEOINT Operations: ~~Tasks, activities, and events necessary to collect, manage, analyze, generate, visualize, and provide imagery, imagery intelligence, and geospatial information necessary to support national and defense missions and international arrangements.~~

~~**GEOINT Standard:** A documented agreement containing technical specifications or other precise criteria to be used consistently as rules, guidelines, or definitions of characteristics to ensure that materials, products, processes, or services are fit for the analysis and visual representation of physical features and geographically referenced activities.~~

Imagery: A likeness or representation of any natural or manmade feature or related object or activity, and the positional data acquired at the same time the likeness or representation was acquired, including products produced by space-based national intelligence reconnaissance systems, and likeness or presentations produced by satellites, airborne platforms, unmanned aerial vehicles, or other similar means. ~~Such~~ ~~The~~ term does not include handheld or clandestine photography taken by, or on behalf of, human intelligence collection organizations. (Reference q) ~~(Title 10 U.S. Code §467)~~

Imagery and Gridded Data Sources: Sensor systems, including associated ground stations that operate on surface/ground-based, naval, airborne or space platforms, with the capability to collect GEOINT data. Sensor phenomenologies include electro-optical, infrared, multispectral, hyperspectral, synthetic aperture radar, moving target indicator, light detection and ranging, and other remote sensing technologies. These systems also include GEOINT data processing systems operated to enable discovery, dissemination, or data processing and transformation services; and systems that produce imagery and gridded foundation data derived from sensor collected data.

Imagery Intelligence: The technical, geographic, and intelligence information derived through the interpretation or analysis of imagery and collateral materials. (Reference q) ~~as defined by Title 10 U.S. Code §467.~~

~~**Independent Test Organization:** A test organization that is independent, both managerially and financially, from the developer whose activities are being evaluated, and staff members performing test and evaluation tasks are not the same staff members as those performing development tasks. [DoD Test and Evaluation Management Guide (TEMG)]~~

~~**Information Environment:** The aggregate of individuals, organizations, and systems that collect, process, disseminate, or act on information. See also information system. (Joint Publication 1-02, DOD Dictionary of Military and Associated Terms, 08 November 2010, as amended through 15 August 2012)~~

Information Needs: A condition or situation requiring knowledge or intelligence derived from received, stored, or processed facts and data.

~~**Information Services Capability:** An IT capability designed to provide awareness of, access to, and delivery of data or information made available for consumption by one or more users. Users can be an individual, organization, or machine. (DoDI 8320.02)~~

Information Support Plan: An information set supporting interoperability test and certification. It identifies and documents information needs, infrastructure support, and IT and NSS interface requirements and dependencies focusing on net-centric, interoperability, supportability, and sufficiency concerns. It is a requirement for all Information Technology (IT) programs, including National Security Systems (NSS), that connect in any way to the communications and information infrastructure. (Reference m) ~~The identification and documentation of information needs, infrastructure support, IT interface requirements and dependencies focusing on net-centric, interoperability, supportability, and sufficiency concerns. (DoDI 4630.8)~~

Information Systems (IS): A discrete set of information resources organized for the collection, processing, maintenance, use, sharing, dissemination, or disposition of information. (Reference f)

Information Technology (IT)/~~Information System (IS):~~ 1. With respect to an executive agency means any ~~Any~~ equipment or interconnected system or subsystem of equipment, used in the automatic acquisition, storage, ~~analysis, evaluation,~~ manipulation, management, movement, control, display, switching, interchange, transmission, or reception of data or information by the executive agency, if the equipment is used by the executive agency directly or used by a contractor under a contract with the executive agency that requires the use - (i) of that equipment; or (ii) of that equipment to a significant extent in the performance of a service or the furnishing of a product; 2. Includes ~~“Equipment” includes~~ computers, ancillary equipment (including imaging peripherals, input, output, and storage devices necessary for security and surveillance), peripheral equipment designed to be controlled by the sentral processing unit of a computer, software, firmware, and similar procedures, services, (including support services), and related resources, 3. ~~Does but IT does~~ not include any equipment required by a federal contractor incidental to a federal contract. (Reference f) ~~(DTM 11-009 (Acquisition Policy for Defense Business Systems (DBS))).~~

For the purposes of this Manual, IT includes: NSS, IT acquisition programs, information systems, IT initiatives, IT services, software, electronic warfare devices, DBS, qualified prototypes, Commercial-Off-the-Shelf (COTS), Government Off-the-Shelf (GOTS), Rapid Acquisition, Special Access Program (SAP), Joint Capability Technology Demonstration (JCTD), Coalition Warrior Interoperability Demonstration, Combatant Command Initiatives Fund (CCIF), and non-program of record materiel solution efforts. ~~(Reference o).~~ The term “information system” is used synonymously with the term “IT” (to include National Security Systems).

Intelligence Certification: The objective of intelligence certification is to ensure intelligence requirements have been identified at the earliest possible point, and that all likely intelligence support requirements and shortfalls (if applicable) have been documented. Threat assessment accomplished as part of intelligence certification further ensures that continuous system and protection threat analysis of applicable adversary threat capabilities is complete and incorporated into capability requirement documents. (Reference g)

Intelligence Supportability: Intelligence support requirements will be reviewed by subject matter experts from DoD and Service intelligence organizations for supportability prior to granting intelligence certification. Joint Staff J-283 /Intelligence Requirements Certification Office (J283/IRCO) provides intelligence certification in accordance with Appendix I to Enclosure F of this manual. (Reference g)

Interface: The functional and physical characteristics required to exist at a common boundary or connection between people, between systems, or between people and systems. Interfaces include all systems external to the system being analyzed that provides a common boundary or service that is necessary for the system being analyzed to perform its mission in a non-degraded mode; e.g., a system that supplies power, cooling, heating, input signals, etc. (Reference m) ~~(DAU-Glossary-of-Defense Acquisition Acronyms and Terms)~~

Interoperability: Denotes the ability to operate in synergy in the execution of assigned tasks. The condition achieved among communications-electronics systems or items of communications-electronics equipment when information or services can be exchanged directly and satisfactorily between them and/or their users. The degree of interoperability should be defined when referring to specific cases. (Reference h) ~~Joint Publication 1-02)~~

For IT (and NSS), interoperability is the ability of systems, units or forces to provide data, information, materiel and services to, and accept the same from, other systems, units or forces, and to use the data, information, materiel, and services so exchanged to enable them to operate effectively together. IT interoperability includes both the technical exchange of information and the ~~end to end~~ operational effectiveness of that exchanged ~~of~~ information as required for mission accomplishment. Interoperability is more than just information exchange. It includes systems, processes, procedures, organizations, and missions over the lifecycle and must be balanced with ~~cybersecurity~~ (formerly ~~Information Assurance~~-(IA)). (Reference f e)

Joint: Connotes activities, operations, organizations, etc., in which elements of two or more Military Departments participate. (Reference h) ~~Joint Publication 1-02)~~

Joint Capabilities Integration and Development System (JCIDS): ~~The Primary system used by the JROC and its subordinate boards to fulfill the CJCS's statutory responsibilities in assessing joint military capabilities, and identifying, approving, and prioritizing gaps in such capabilities, to meet applicable requirements in the NDS [National Defense Strategy]. (Reference m) A Chairman of the Joint Chiefs of Staff process to identify, assess, and prioritize joint military capability needs. The JCIDS process is a collaborative effort that uses joint concepts and DOD Information Enterprise Architecture and solution architectures to identify prioritized capability gaps and integrated DOTMLPF solutions (materiel and non-materiel) to resolve those gaps. (CJCSI 3170.01 Series)~~

Joint Capability Area: Collections of like DoD activities functionally grouped to support capability analysis, strategy development, investment decision making, capability

portfolio management, and capabilities-based force development and operational planning. (Reference m) ~~(https://intellipedia.intelink.gov/wiki/Joint_Capability_Areas)~~

Joint Capability Technology Demonstration (JCTD): A demonstration of the military utility of a significant new technology, and an assessment to clearly establish operational utility and system integrity. (Reference m)

Joint Interface: An Information Technology (IT) interface that passes, or is used to pass, information between systems and equipment operated by two or more Combatant Commanders (CCDRs), Services, or agencies. (Reference m)

Joint Interoperability Certification: Certification issued only by JITC that involves an evaluation of information interoperability with respect to interoperability requirements at the joint level. JITC updates interoperability certifications throughout a system's life cycle to reflect changes in the system, status, and joint interoperability environment. All JITC interoperability certifications expire upon changes that may affect interoperability. (Reference o) ~~(JITC Interoperability Process Guide)~~

~~**Joint Interface:** An IT interface that passes, or is used to pass, information between systems and equipment operated by two or more combatant commanders, Services, or agencies.~~

Joint Mission Thread: An operational and technical description of the end-to-end set of activities and systems that accomplish the execution of a joint mission. (Reference m)

Key Performance Parameters (KPPs): These are the performance attributes of a system considered critical or essential to the development of an effective military capability. Failure of a system to meet a validated KPP threshold value triggers a review by the validation authority and evaluation of operational risk and/or military utility of the associated system(s). The review may result in validation of an updated KPP threshold value modification of production increments, or recommendation for program cancellation. (Reference f) ~~Those capabilities or characteristics considered essential for successful mission accomplishment. Failure to meet a system's or program's KPP threshold can be cause for the concept or system selection to be reevaluated or the program to be reassessed or terminated. Failure to meet a system or program's KPP threshold can be cause for the family-of-systems or system-of-systems concept to be reassessed, or the contributions of the individual systems to be reassessed. KPPs are validated by the JROC. KPPs are included in the acquisition program baseline. (JCIDS Manual)~~

Legacy System: Any fielded system scheduled for replacement, integration, or to be subsumed, that must be maintained, but will not be modified.

Measures of Effectiveness: An indicator used to measure a current system state, with change indicated by comparing multiple observations over time. ~~A criterion used to assess changes in system behavior, capability, or operational environment that is tied to~~

~~measuring the attainment of an end state, achievement of an objective, or creation of an effect.~~ Also called MOE. (Reference h) (DoD JP 1-02)

Measures of Performance: An indicator used to measure a friendly action that is tied to measuring task accomplishment. ~~A criterion used to assess friendly actions that is tied to measuring task accomplishment.~~ Also called MOP. (Reference h) (DoD JP 1-02)

~~**Milestone Decision Authority (MDA):** The individual designated in accordance with criteria established by the USD (AT&L), or by the DoD CIO for acquisition programs, to approve entry of an acquisition program into the next phase. (DoDI 5000.02) The MDA for IT that involves equipment that is an integral part of a weapon or weapon system, or is an acquisition of services program is, or will be, designated by the USD (AT&L).~~

Mission Area: A defined area of responsibility with functions and processes that contribute to mission accomplishment. (Reference s)

National Security Systems (NSS): Information system (including any telecommunications system) used or operated by an agency or by a contractor of an agency, or other organization on behalf of an agency, the function, operation, or use of which (1) involves intelligence activities; (2) involves cryptologic activities related to national security; (3) involves the command and control of military forces; (4) involves equipment that is an integral part of a weapon or weapons systems; or (5) is critical to the direct fulfillment of military or intelligence missions. Subsection (5) in the preceding sentence does not include procurement of automatic data processing equipment or services to be used for routine administrative and business applications (including payroll, finance, logistics, and personnel management applications). NSS include any information system (including any telecommunications system) protected at all times by procedures established for information that have been specifically authorized under criteria established by an Executive order or an Act of Congress to be kept classified in the interest of national defense or foreign policy. (Reference d) (DoDI 8330.01)

National System for Geospatial Intelligence (NSG): The combination of technology, policies, capabilities, doctrine, activities, people, data, and communities necessary to conduct Geospatial Intelligence in a multi-intelligence, multi-domain environment. (Reference k) (NSG-Pub 1-0)

~~**NSG Enterprise:** The “functional GEOINT subset” of the DoD Information Enterprise Architecture (DoD IEA) for the Department of the Defense, the Intelligence Community Information Technology Enterprise (IC ITE), and equivalent concepts for the other elements of the NSG that use or provide any capability or service enabled by imagery, imagery intelligence, or geospatial information. By definition each of the systems of the GE is also a member of a larger enterprise. (Reference u) (NSGD 3201)~~

~~**Net-Centric:** Information-based operations that use service-oriented information processing, networks, and data from the following perspectives: user functionality (capability to adaptively perform assigned operational roles with increasing use of nand loosely coupled services), and enterprise management (net operations). The ability to~~

~~provide a framework for full human and technical connectivity and interoperability that allows all DOD users and mission partners to share the information they need, when they need it, in a form they can understand and act on with confidence, and protects information from those who should not have it.~~

Net-Ready Performance Attribute: The net-ready performance attribute ensures interoperability between individually developed and fielded capability solutions. (Reference f)

Net-Ready: ~~DoD IT that meets required information needs, information timeliness requirements, has IA accreditation, and meets the attributes required to support military operations, to be entered and managed on the network, and to effectively exchange information for both the technical exchange of information and the operational effectiveness of that exchange. DOD IT that is net-ready enables warfighters and DOD business operators to exercise control over enterprise information and services through a loosely coupled, distributed infrastructure that leverages service modularity, multimedia connectivity, metadata, and collaboration to provide an environment that promotes unifying actions among all participants. Net readiness requires that IT operate in an environment where there exists a distributed information processing environment in which applications are integrated; applications and data independent of hardware are integrated; information transfer capabilities exist to ensure communications within and across diverse media; information is in a common format with a common meaning; there exist common human-computer interfaces for users; and there exists effective means to protect the information. Net-Readiness is critical to achieving the envisioned objective of a cost-effective integrated environment. Achieving and maintaining this vision requires interoperability:~~

- ~~a. Within a Joint Task Force/combatant command area of responsibility (AOR).~~
- ~~b. Across combatant command AOR boundaries.~~
- ~~c. Between strategic and tactical systems.~~
- ~~d. Within and across Services and agencies.~~
- ~~e. From the battlefield to the sustaining base.~~
- ~~f. Among U.S., Allied, and Coalition forces.~~
- ~~g. Across current and future systems. (Reference f o)~~

N-R KPP Attributes: ~~The N-R KPP Description includes three attributes, the system specific details for which are used to determine whether a system satisfies the N-R KPP. These attributes are: support net-centric military operations, enter and be managed in the network, and exchange information. (Reference p)~~

Net-Ready Key Performance Parameter (N-R KPP): ~~The N-R KPP documents sponsor identified and JROC validated verifiable performance measures and metrics for interoperability engineering, design, and testing. To meet N-R KPP attributes, IT must be able to support military operations, to be entered and managed on the network, and~~

~~to effectively exchange information. The N-R KPP development process will help verify operationally effective provider-to-consumer, end-to-end information exchanges according to the sponsor's stated capability requirements and applicable reference models and reference architectures. It informs the solution architecture according to the DOD Information Enterprise Architecture (IEA). (Reference g p)~~

~~*Note: DoD Instruction 8330.01 has since separated validation authority based on the presence of joint, multinational or interagency interoperability requirements: "Or the appropriate DoD Component for all IT with no joint, multinational, or interagency interoperability requirements as delegated by the CJCS." This directive and its guiding manuals reflect that bifurcation.~~

~~The N-R KPP assesses information requirements, information timeliness, and net ready attributes are required for both the technical exchange of information and the end-to-end operational effectiveness of that exchange. The N-R KPP consists of measurable and testable performance measures and associated metrics required to evaluate the timely, accurate, and complete exchange and use of information to satisfy information needs for a given system. (Reference d) (DoDI 8330.01)~~

~~**N-R KPP Attributes:** The N-R KPP Description includes three attributes, the system specific details for which are used to determine whether a system satisfies the N-R KPP. These attributes are: support net-centric military operations, enter and be managed in the network, and exchange information. (Reference g)~~

~~**N-R KPP Certification:** An authoritative act or process of supporting or corroborating whether IT interoperability requirements are appropriate and complete. (DoDI 8330.01)~~

~~**N-R KPP Certification Authority:** The office with the authority to certify the N-R KPP; verifies that the system sponsor has properly scoped, refined, and justified the interoperability requirements of the system. The CJCS is the N-R KPP Certification Authority and may delegate this authority to the appropriate DoD Component for all IT with no joint, multinational, or interagency interoperability requirements. (Reference d) (DoDI 8330.01)~~

~~**N-R KPP Effectiveness and Performance Measures:** The portion of the N-R KPP that describes the measurable and testable Operational Requirements for the N-R KPP. These Operational Requirements are the Threshold and Objective performance values for each of the N-R KPP Attributes. The full description from the N-R KPP Compliance Statement is as follows: The capability, system, and/or service and/or service must fully support execution of joint critical operational activities and information exchanges identified in the DOD Enterprise Architecture and solution architectures based on integrated DODAF content. (Reference g p)~~

NSG Enterprise: The IT infrastructure for delivering GEOINT data and services across the NSG. The "functional GEOINT subset" of the DoD Information Enterprise Architecture (DoD IEA) for the Department of the Defense, the Intelligence Community Information Technology Enterprise (IC ITE), and equivalent concepts for the other elements of the NSG that use or provide any capability or service enabled by imagery,

imagery intelligence, or geospatial information. By definition each of the systems of the GE is also a member of a larger enterprise. (Reference j) (NSGD-3201)

NSG Fora: Established NSG Fora are the NSG Senior Management Council (NSMC) and the National Geospatial Intelligence Committee (GEOCOM).

Chaired by the GEOINT Functional Manager, the NSMC is the senior, three-star equivalent functional management advisory council of the NSG. Members discuss strategic, community-wide GEOINT issues and jointly address areas of common concern. The GEOINT FM brings a variety of issues before the NSMC for review, deliberation, and/or decision. (Reference i) (NSGD-1100)

The mission of the GEOCOM is to assist and support the GEOINT FM in discharging his/her obligations and authorities and to serve as the National GEOINT Committee of Record. The GEOCOM provides a forum for GEOINT consumers and producers to exchange information; promote integration; synchronize GEOINT resource programs, investments, capabilities, and priorities; and support NSMC activities. (Reference i) (NSGD-1100)

Performance Testing: Any test activity which evaluates whether system output data and service interfaces are sufficiently compliant with standards, and have the full richness of metadata to enable the operationally effective transport of the data and services across the Global Information Grid (GIG) so that any user of the data and/or service has the opportunity to fully exploit the information content to the extent of the capability enabled by the system configuration design. Performance testing applies GEOINT-specific measurable and testable criteria for interoperability, and for operationally effective end-to-end information exchanges, based on threshold and objective values that determine the data and/or service's measures of effectiveness (MOE) and measures of performance (MOP).

Product Specification: A detailed description of a dataset, together with additional information about the dataset, that will enable it to be created, supplied to, and used by another party.

Program of Record: IT with a program element funded through the program objective memorandum process and included in the FYDP.

Register: A formal or official list or recording of items.

Registry: A managed place where data, records, lists, registers, or similar representations of controlled content is kept for reference. The content of a registry is managed by a maintenance authority and may be organized into one or more lists or registers.

Self-Assessment: An assessment performed by the system sponsor or program manager in consonance with the assessment processes and methods established by the GFMSA procedures.

Solution Architecture: A framework or structure that portrays the relationships among all the elements of something that answers a problem. This architecture type is used to define a particular project to create, update, revise, or delete established activities in the Department. ~~Solution architecture may be developed to update or extend another architecture. A solution architecture is the most common type of architecture developed in the Department.~~ (Reference m) (DoDAF V2.0)

Source: Any data resource that contributes to the creation of GEOINT. This includes resources that provide data describing the Earth's natural and manmade features, as well as natural and human activities that can be referenced to specific locations.

~~**Standards Conformance Testing:** Testing performed to determine the extent to which a system or subsystem adheres to, or implements, a standard.~~

Standards Conformance Certification: Confirmation that an IT/IS has undergone IT standards conformance testing with respect to a given standard, and correctly implements the standard with specified profiles and options.

~~**Standards Conformance Testing:** Testing performed to determine the extent to which a system or subsystem adheres to, or implements, a standard.~~

System Performance Requirements: Performance requirements the system must meet in order to satisfy its Operational Requirements.

~~**Test:** Any program or procedure designed to obtain, verify, or provide data for the evaluation of any of the following: progress in accomplishing developmental objectives; the performance, operational capability, and suitability of systems, subsystems, components, and equipment items; and the vulnerability and lethality of systems, subsystems, components, and equipment items. An activity designed to provide data on functional features and equipment operation under fully controlled and traceable conditions. The data are subsequently used to evaluate quantitative characteristics. (Reference m) (DAG)~~

Test and Evaluation Strategy (TES): An early test and evaluation (T&E) planning document that describes (T&E) ~~test and evaluation~~ activities starting with Technology Development (TD) and continuing through Engineering and Manufacturing Development (EMD) and Production and Deployment (P&D). ~~System Development and Demonstration into Production and Deployment.~~ The TES describes how component technologies being developed will be demonstrated in a relevant environment to support the program's transition into the EMD Phase. ~~System Development and Demonstration Phase.~~ Over time, the scope of this document will expand and evolve into the Test and Evaluation Master Plan (TEMP) due at Milestone B. (Reference m)

For the purpose of this document Engineering and Manufacturing Development (EMD) and Production and Deployment (P&D) is synonymous with System Development and Demonstration and Production and Deployment (P&D).

Unanticipated Use: Any use of the data or services described in an architecture which have not previously been defined as an operational use in the ~~Initial Capabilities Document (ICD), Joint DOTmLPP-P Change Recommendation (DCR), Concept of Operations (CONOPS), Capability Development Document (CDD), and Capability Production Document (CPD). ICD, DCR, CONOPS, CDD, and CPD.~~ (Reference m)

Unanticipated Users: Users who do not provide advance indication they will use data or services. (Derived from Reference m)

Validation: Provision of objective evidence that the capability provided by the system complies with stakeholder performance requirements, achieving its use in its intended operational environment. Validation answers the question, “Is it the right solution to the problem?” Validation consists of evaluating the operational effectiveness, operational suitability, sustainability, and survivability of the system or system elements under operationally realistic conditions. (Reference l) (DAG)

Vector Data: A digital representation of static real-world features. Vectors are commonly modeled as a geometry with attributes. The geometry describes the shape and location of the vector while the attributes describe what it is. Vector data is usually captured using the GEOINT Structure Implementation Profile (GSIP) feature model and one of the approved Feature Data Dictionaries.

Verification: Provision of evidence that the system or system element performs its intended functions and meets all performance requirements listed in the system performance specification and in functional and allocated baselines. Verification answers the question, “Did you build the system correctly?” Verification is a key risk-reduction activity in the implementation and integration of a system, and enables the program to identify defects in system elements before integration at the next level, thereby preventing costly troubleshooting and rework. (Reference m) (DAG)

~~**Web Services:** A web-based mechanism to enable access to one or more capabilities, where the access is provided using a prescribed interface and is exercised consistent with constraints and policies as specified by the service description. (DoD NCSS) An IT web service is an IT capability designed to provide awareness of, access to, and delivery of data or information made available for consumption by one or more users. Users can be an individual, organization, or machine. (Reference c) (DoDI 8320.02) GEOINT web services are a self-contained and self-describing means of providing network-enabled applications that communicate using interoperable (open) standards over an Internet Protocol (IP) network with access to GEOINT data and processing services. These web services enable the sharing of business logic, data, and processes through a programmatic interface across a network.~~